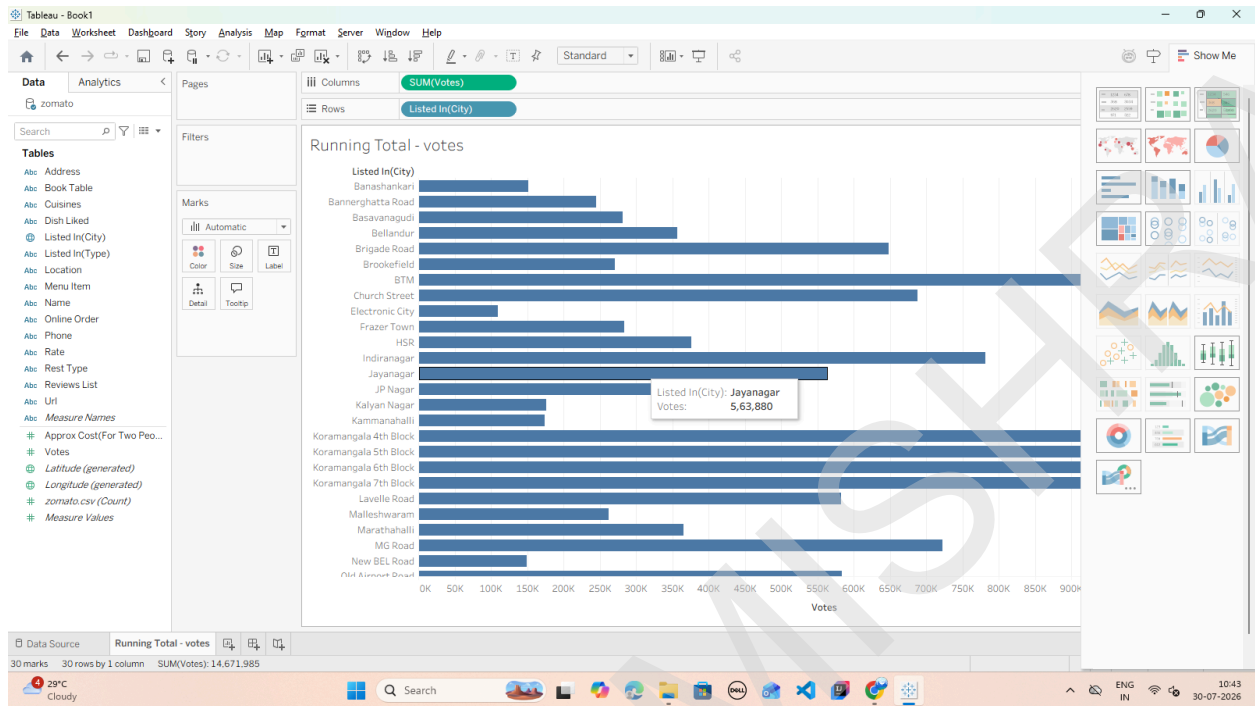


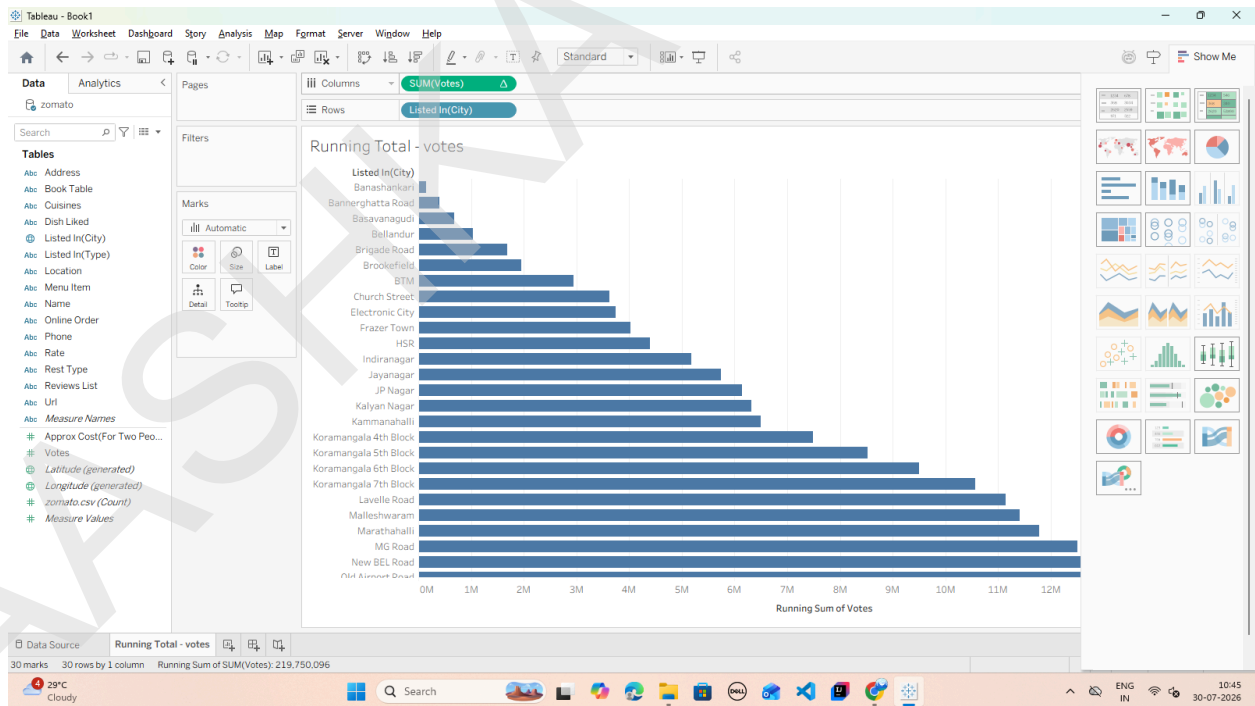


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2. From the Data Pane, drag listed\_in(city) to the Rows shelf.
3. Drag votes to the Columns shelf.



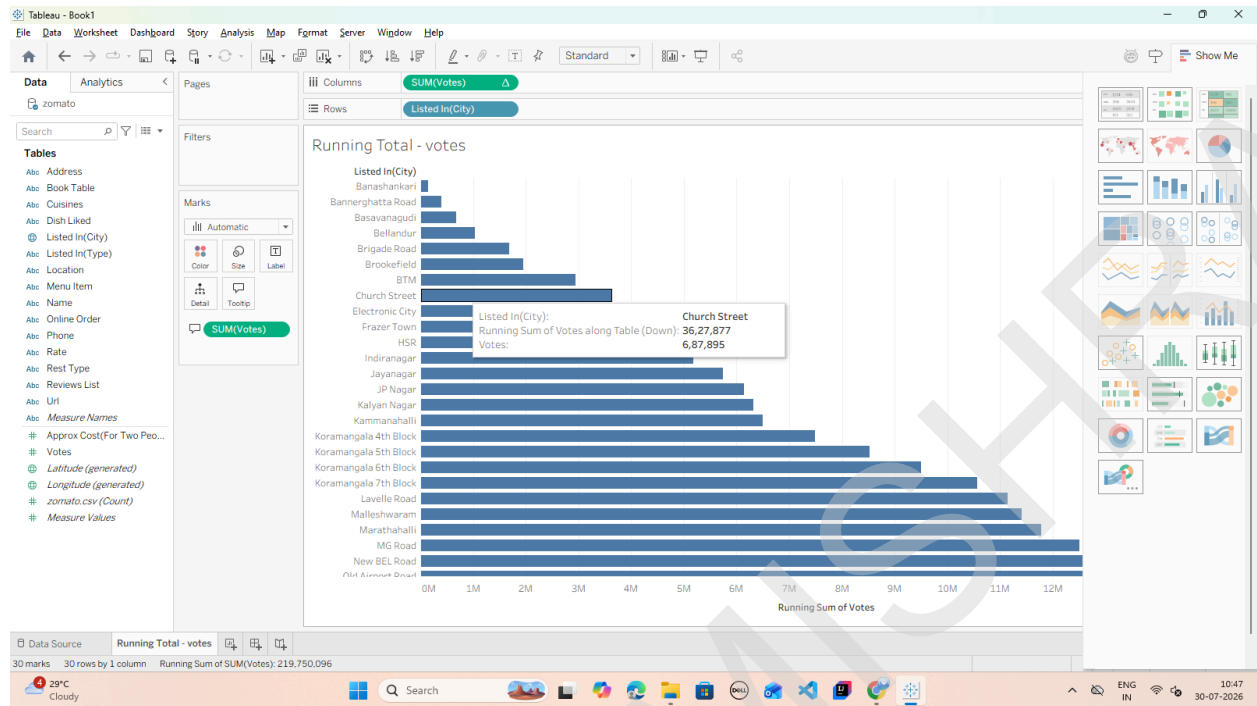
4. Convert to Quick Table Calculation:



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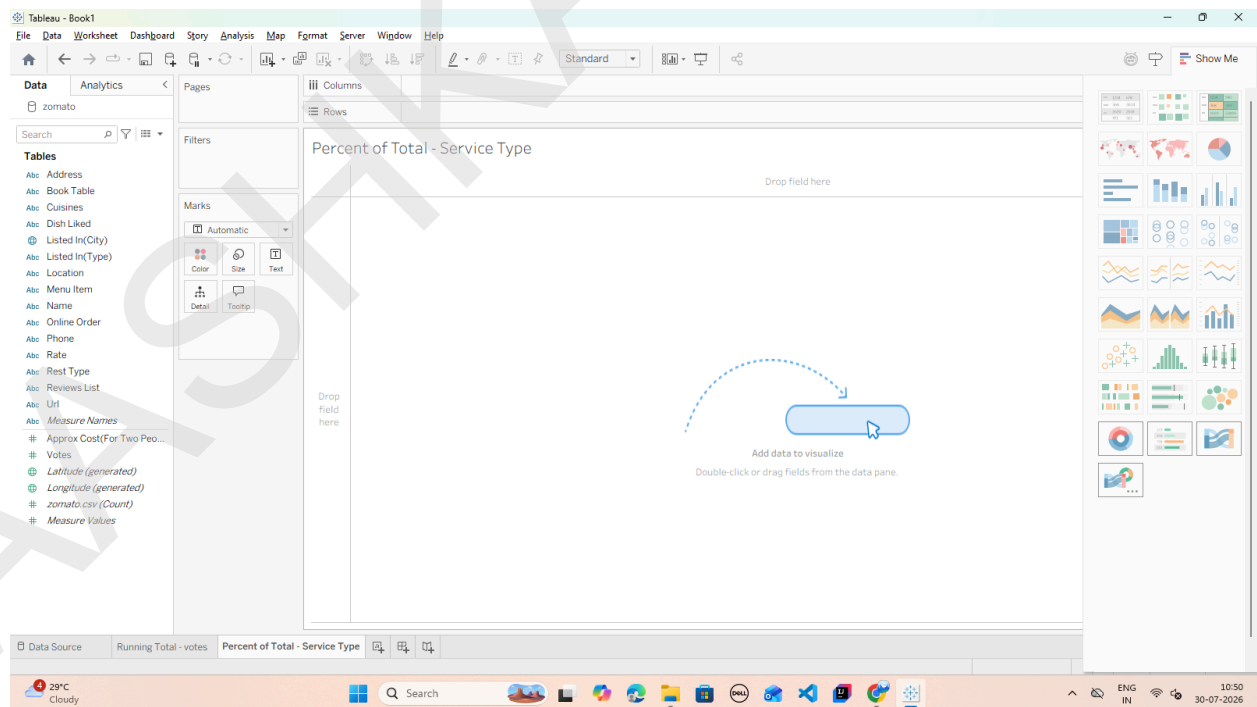
5. Add Visual Context (Secondary Measure):



Task 2: Create a Percent of Total across Categories

A Percent of Total transforms raw numbers into relative percentages of the entire view (summing to 100%).

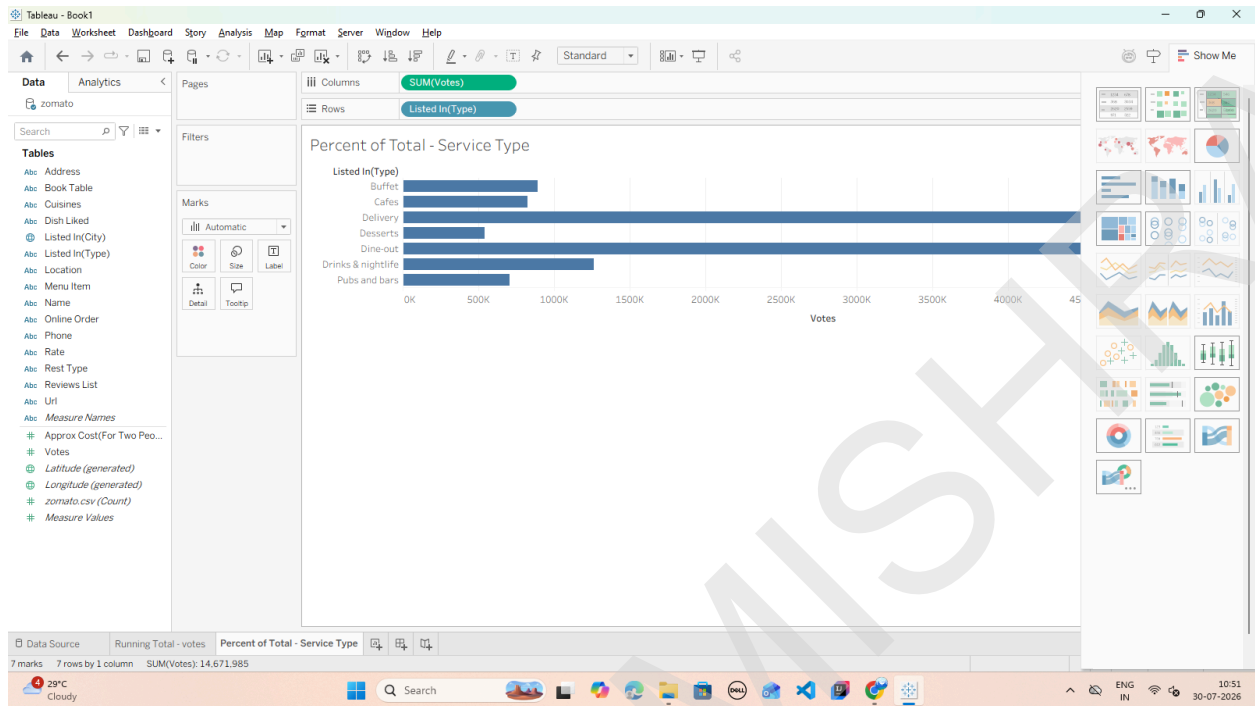
1. Open a new worksheet and rename it: Percent of Total - Service Type.



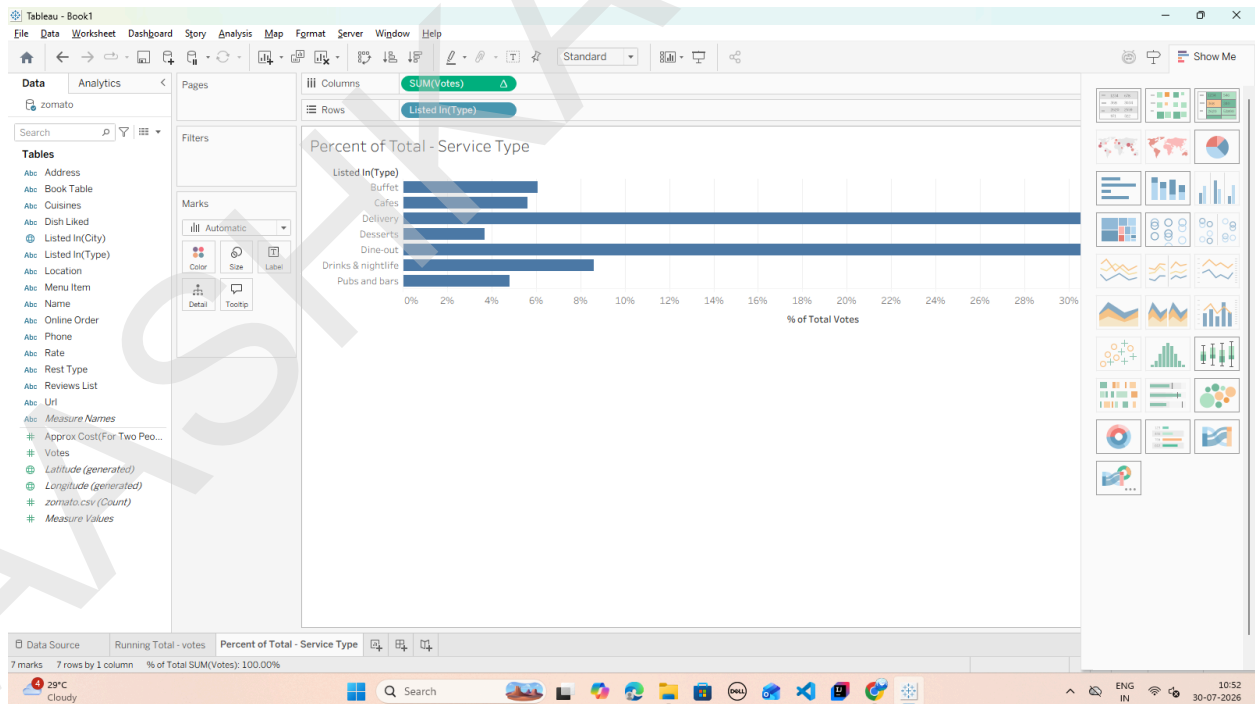
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2. Drag listed\_in(type) (e.g., Buffet, Cafes, Delivery, Dining) to the Rows shelf.
3. Drag votes to the Columns shelf.



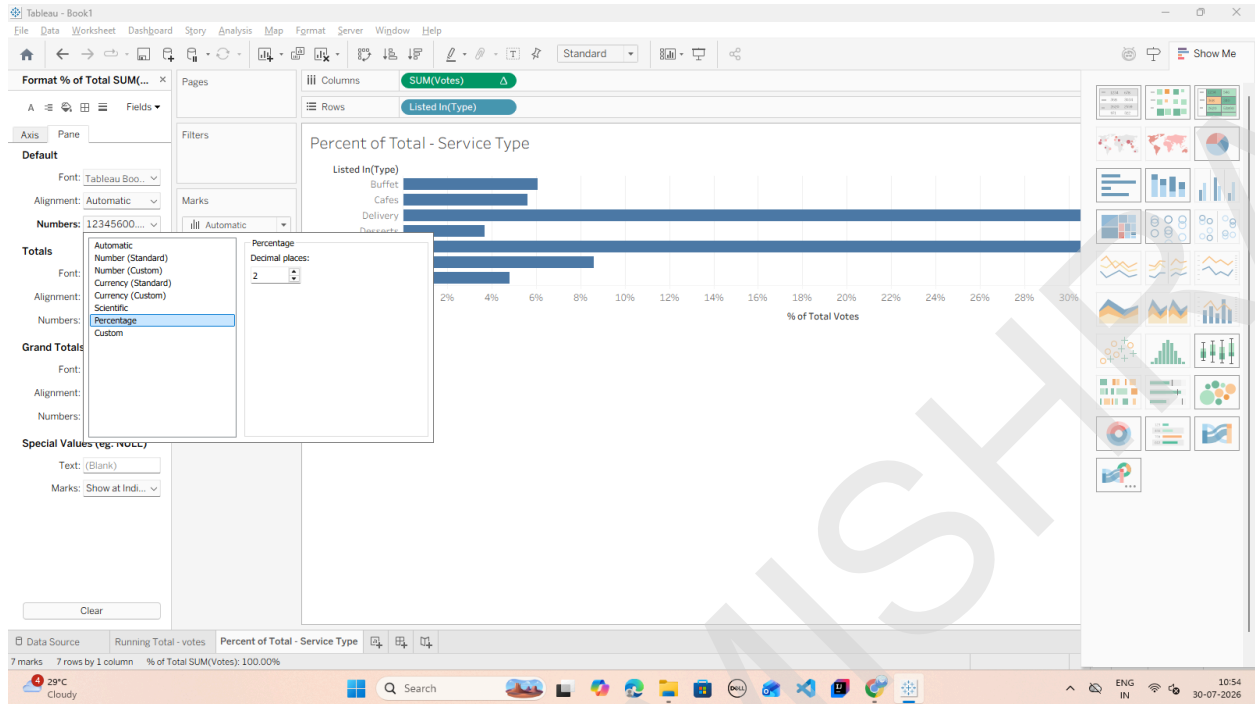
4. Right-click the SUM(votes) pill on the Columns shelf.
5. Hover over Quick Table Calculation and select Percent of Total.



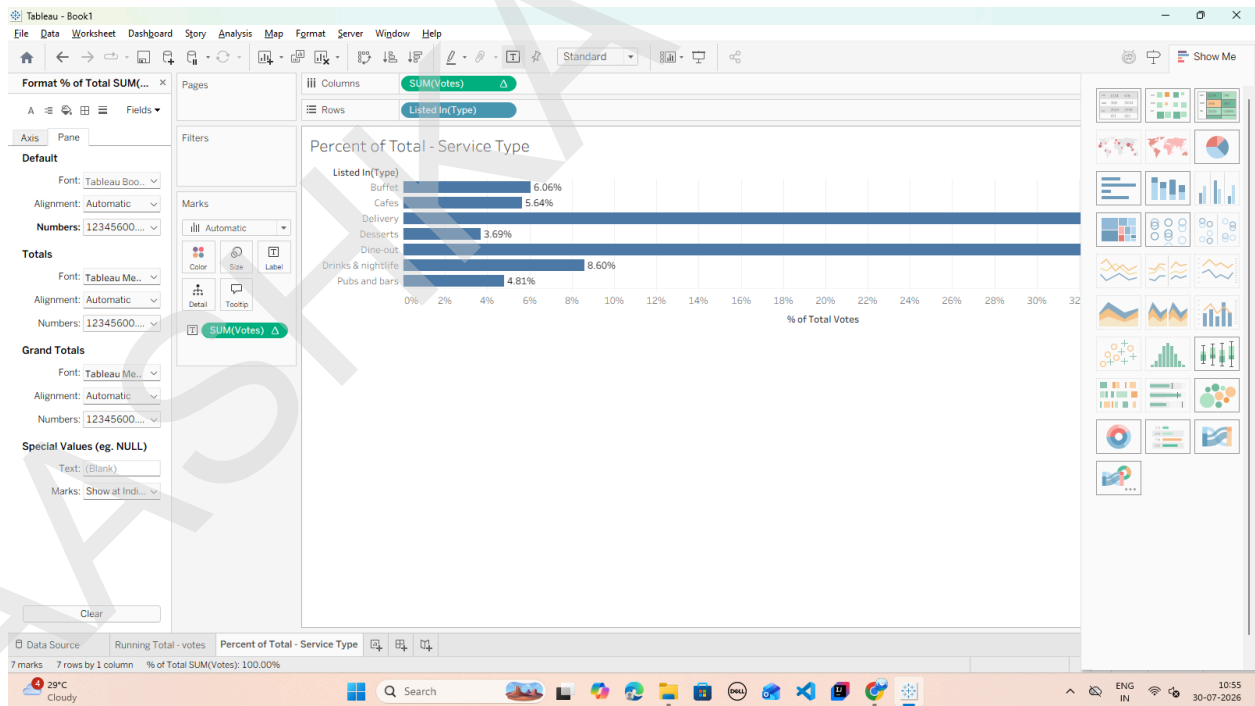
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**6. Format as Percentage:**



**7. Drag SUM(votes) onto the Label mark so the exact percentages display on top of each bar.**



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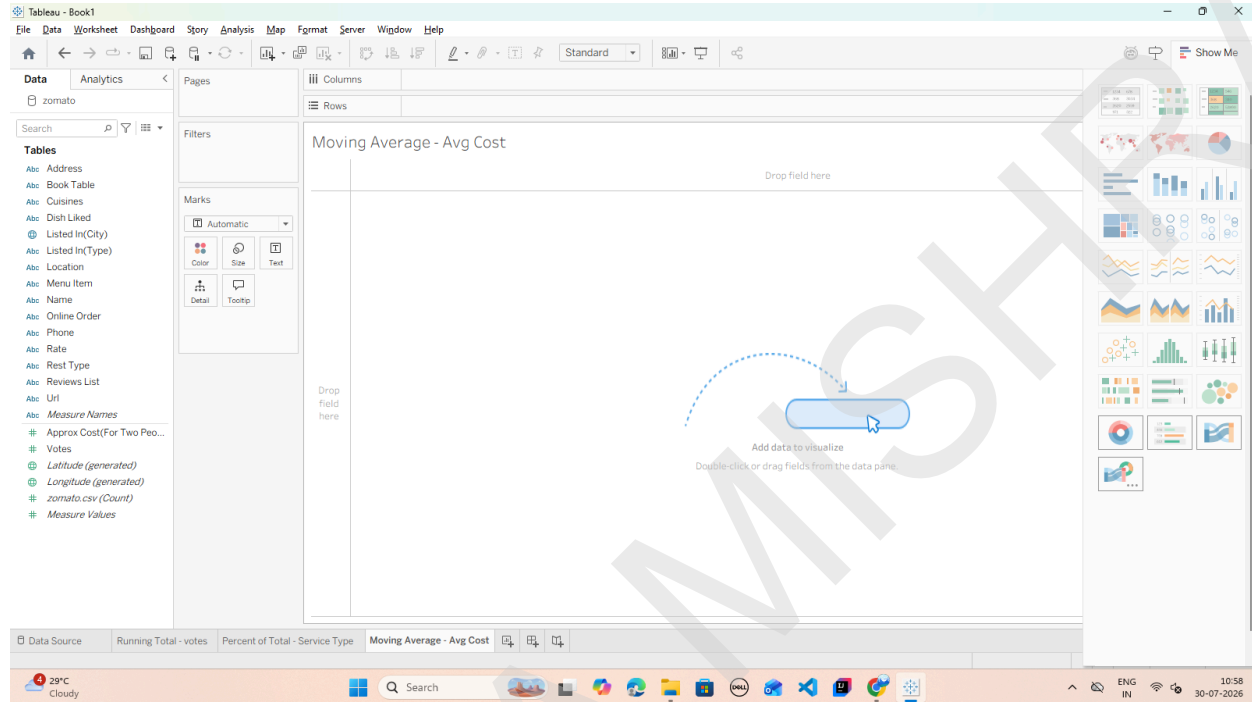
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B: Create moving averages and cumulative metrics

Task 1: Create a Moving Average Calculation

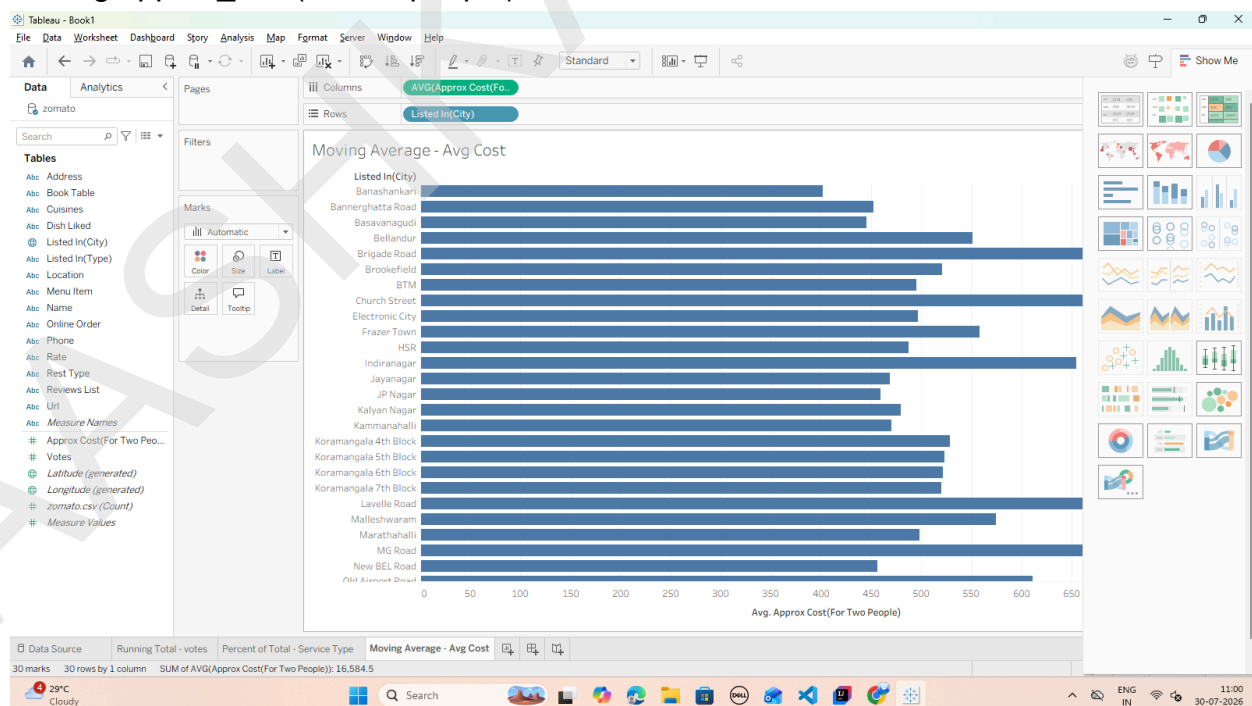
We will create a moving average to smooth out variations in pricing across restaurants based on vote counts or listing order.

1. Open a new worksheet and rename it: Moving Average - Avg Cost.



2. Drag listed\_in(city) to the Rows shelf.

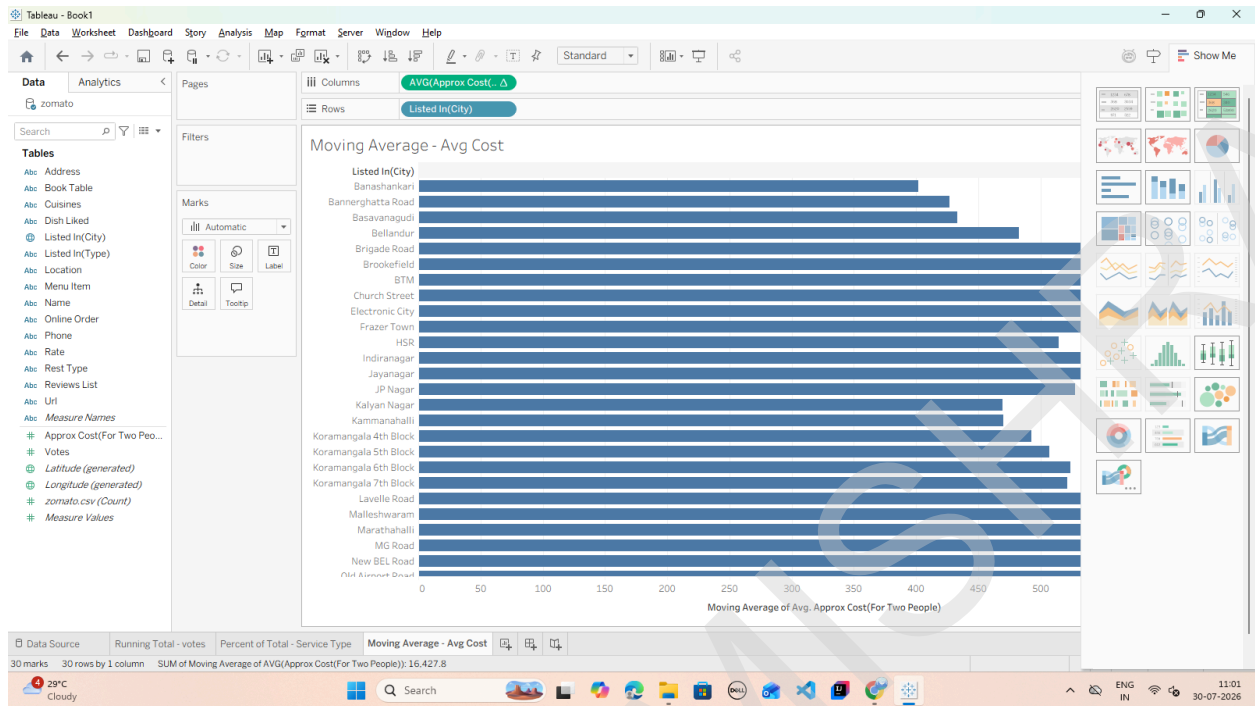
3. Drag approx\_cost(for two people) to the Columns shelf.



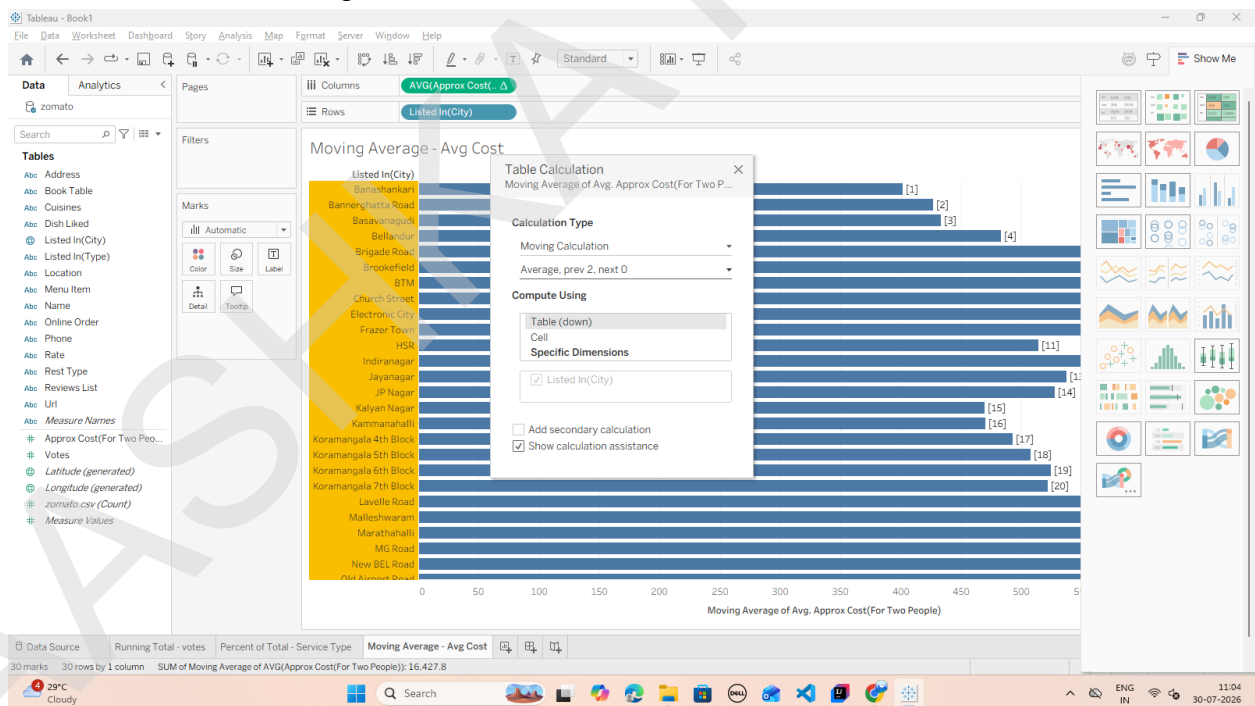
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4. Convert to a Moving Average Quick Table Calculation:



5. Customize the Moving Window:



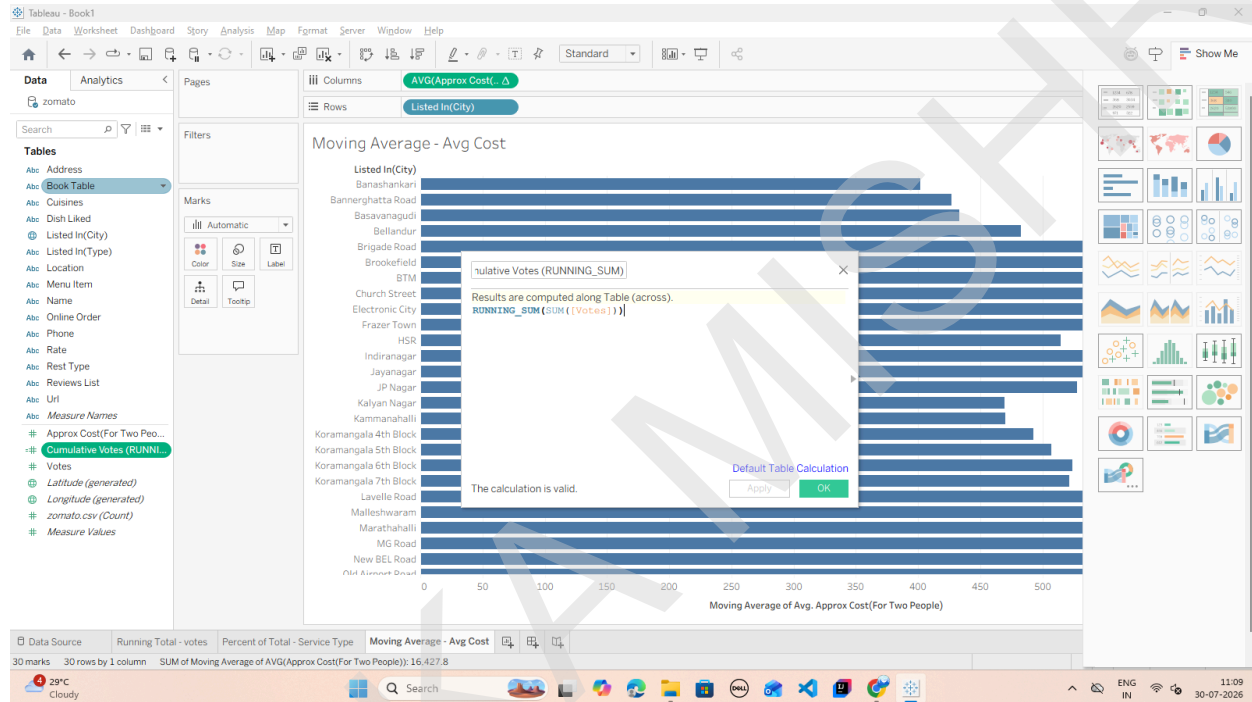
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**Task 2: Create a Custom Cumulative Metric via Calculated Field**

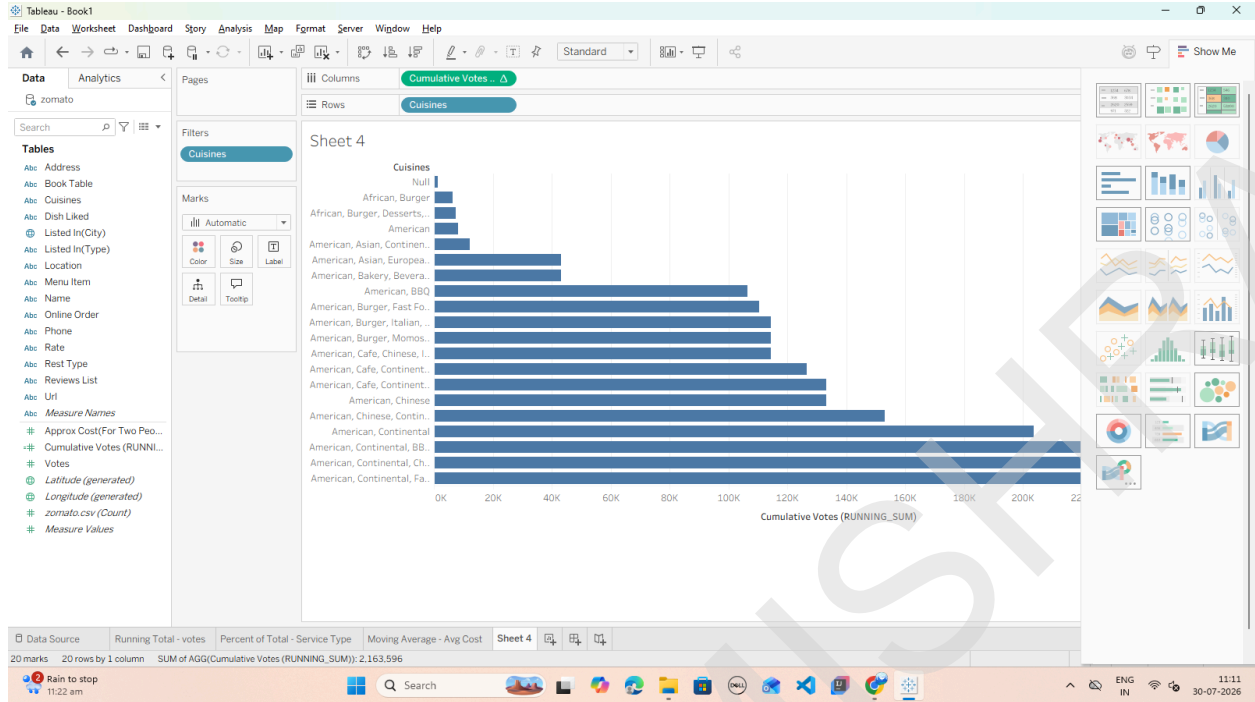
Instead of using Quick Table Calculations, we can write a dedicated Table Calculation function formula directly.

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.
2. Name the field: Cumulative Votes (RUNNING\_SUM).
3. Enter the following Table Calculation formula: `RUNNING_SUM(SUM([votes]))`



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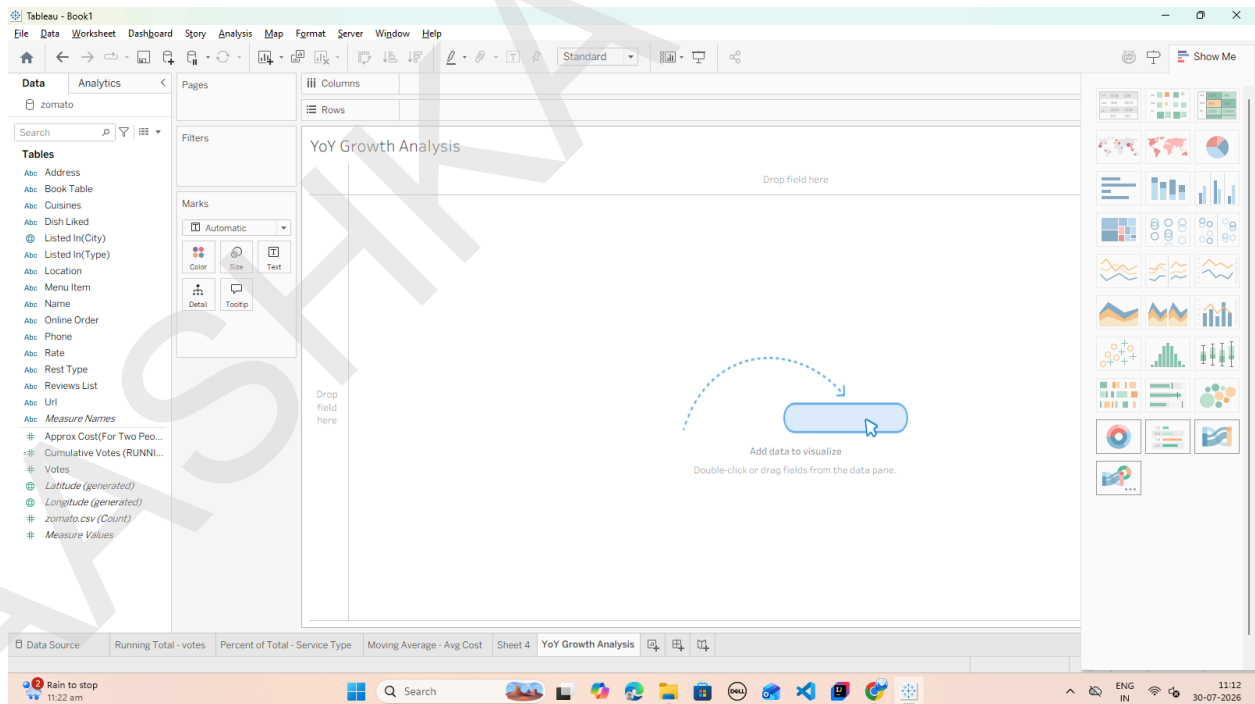
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C: Perform year-over-year growth analysis

Task: Calculate Year-over-Year (YoY) / Period-over-Period Percent Growth

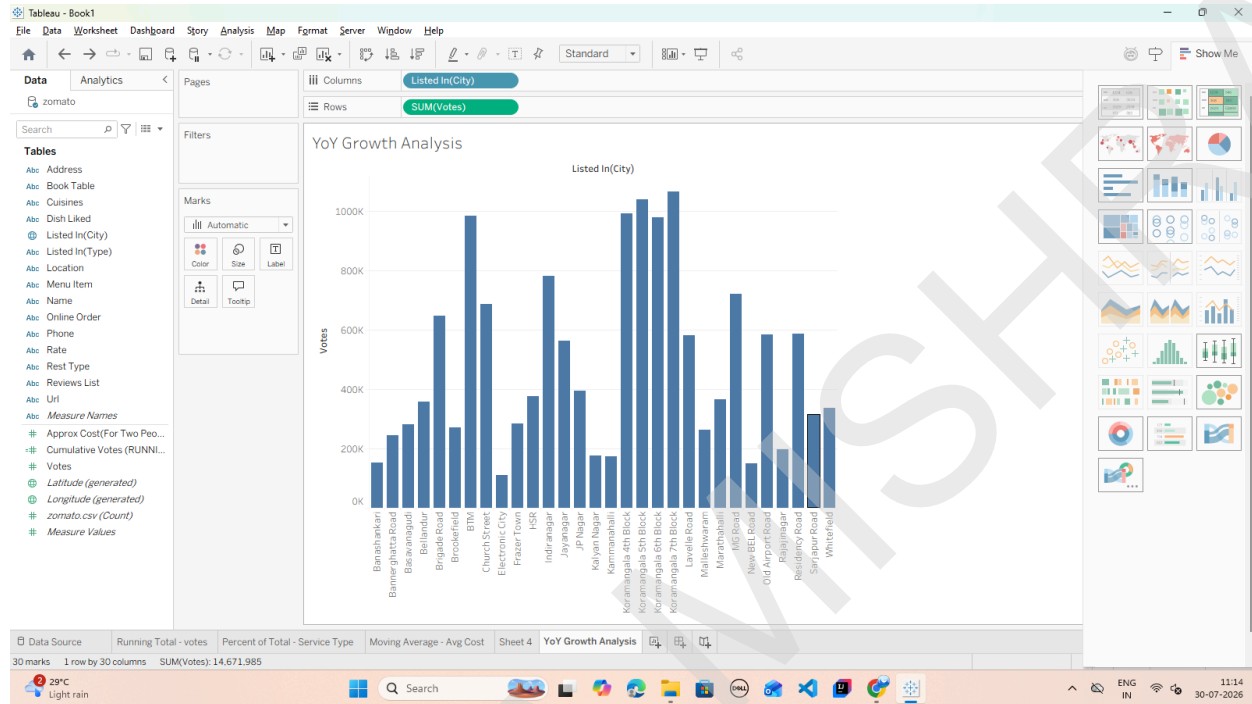
1. Open a new worksheet and rename it: YoY Growth Analysis.



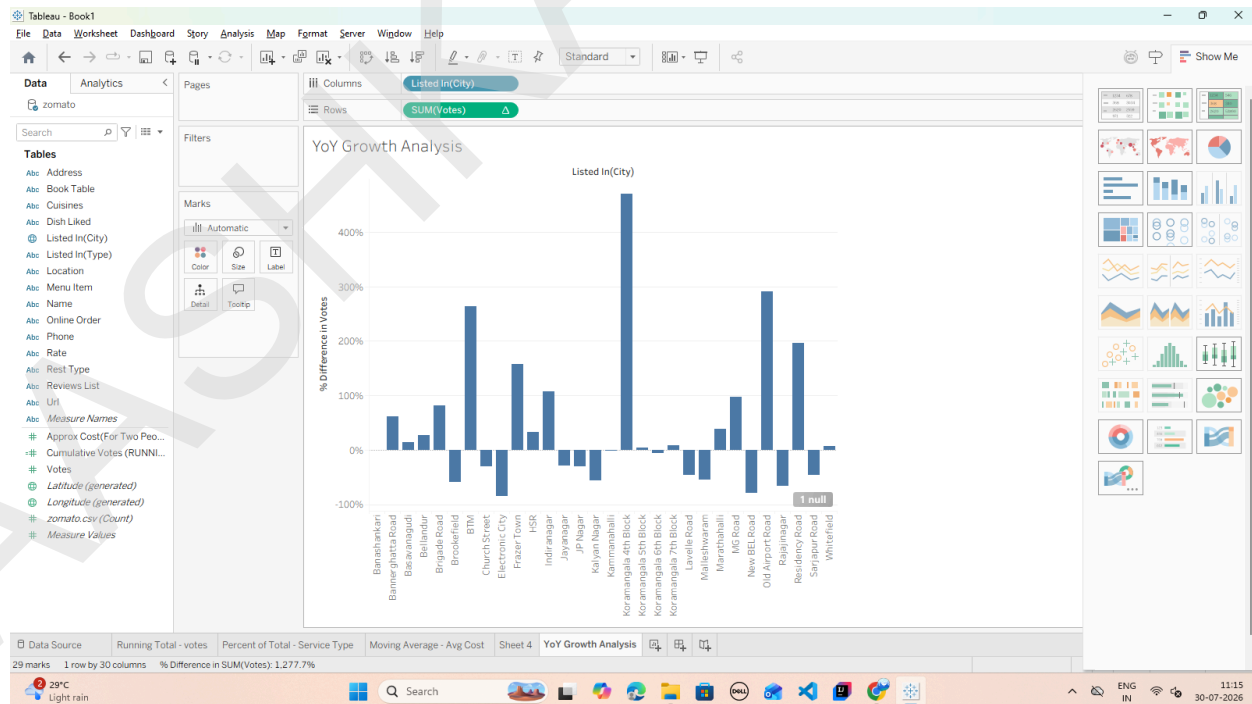
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2. Place your time dimension (or ordinal grouped column like listed\_in(city)) on the Columns shelf.
3. Drag votes (or approx\_cost(for two people)) onto the Rows shelf as SUM(votes).



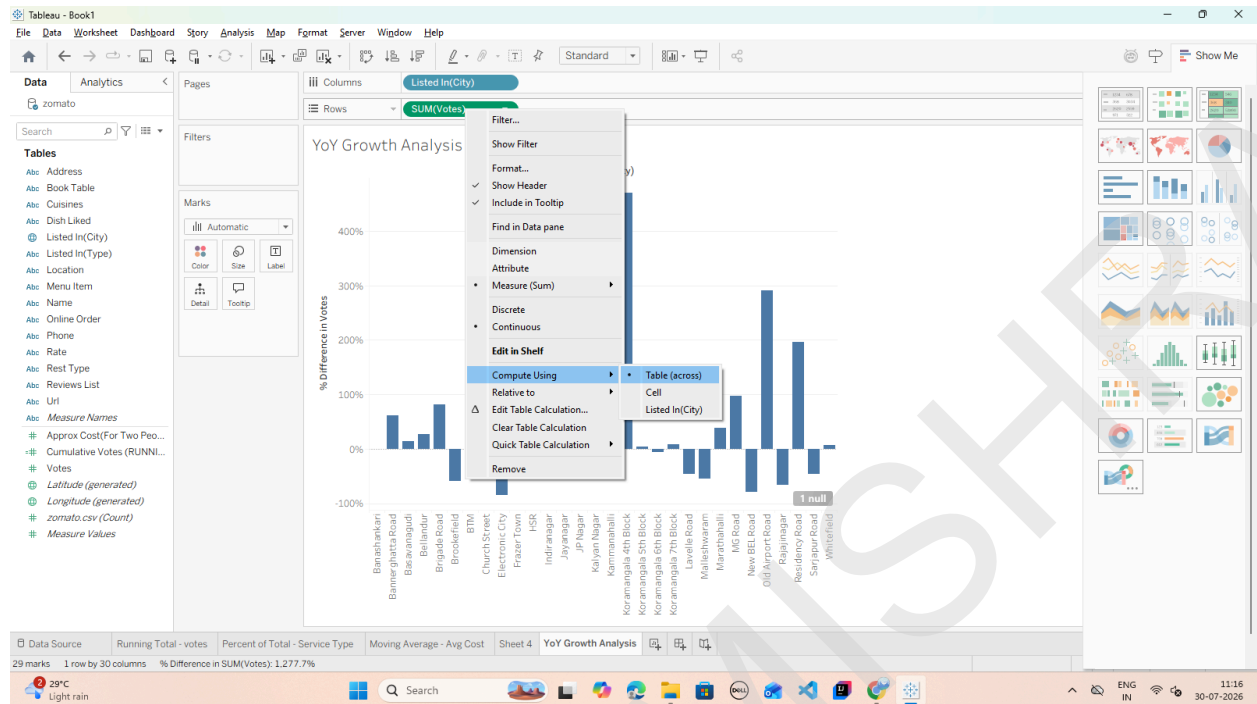
4. Apply the Year-over-Year Growth Quick Table Calculation:



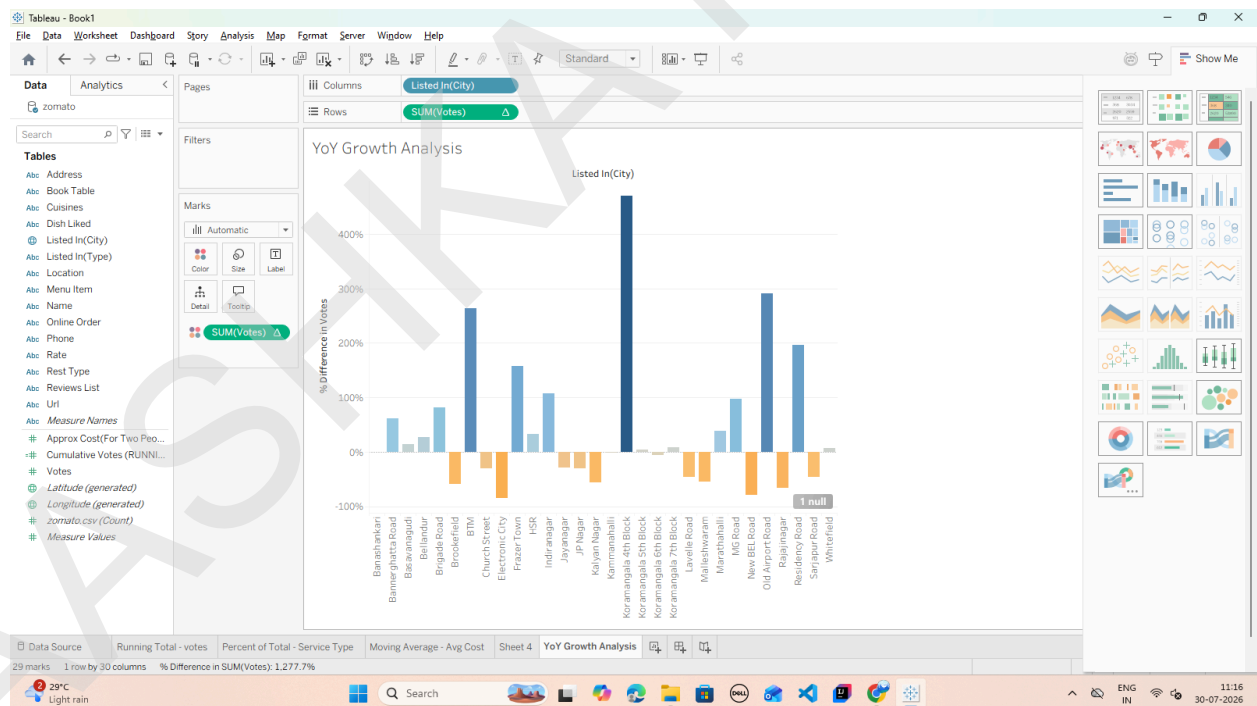
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5. Configure the Growth Calculation Direction:



6. Add Visual Color Indicators:



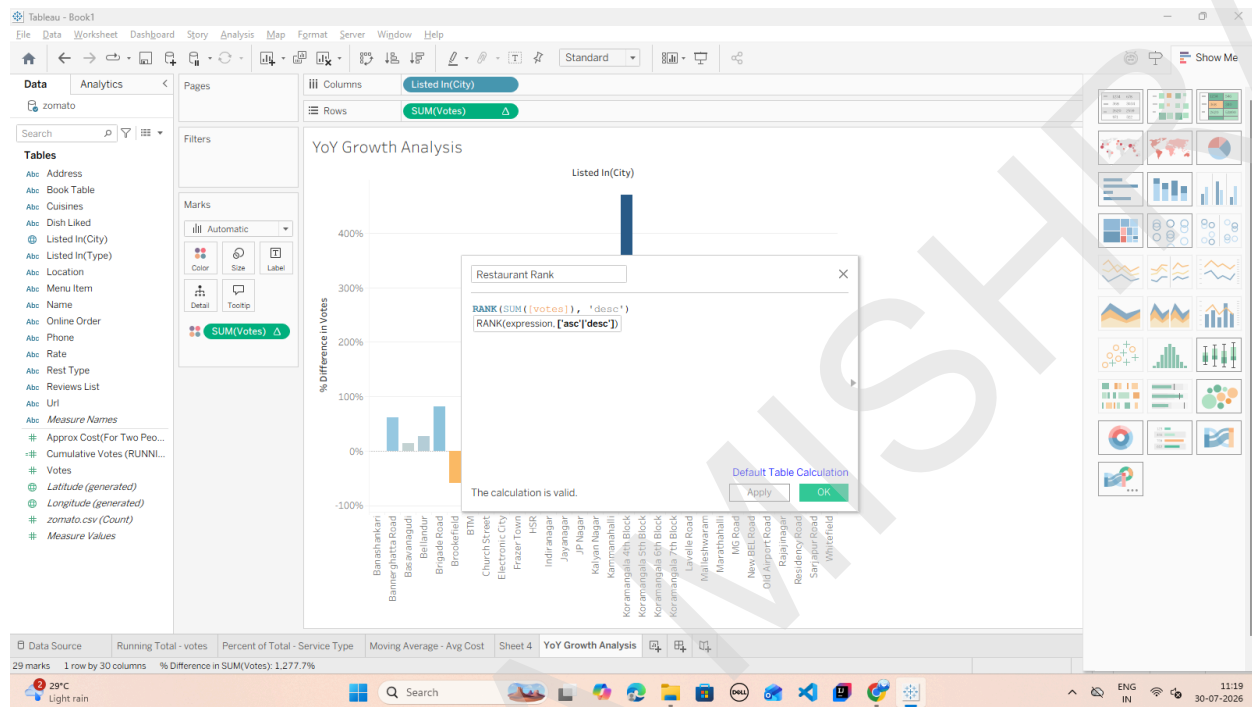
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D: Use ranking functions (Top N, Bottom N) Ranking table calculations evaluate each row's aggregate value relative to all other rows in the partition, assigning a numeric rank (1, 2, 3...).

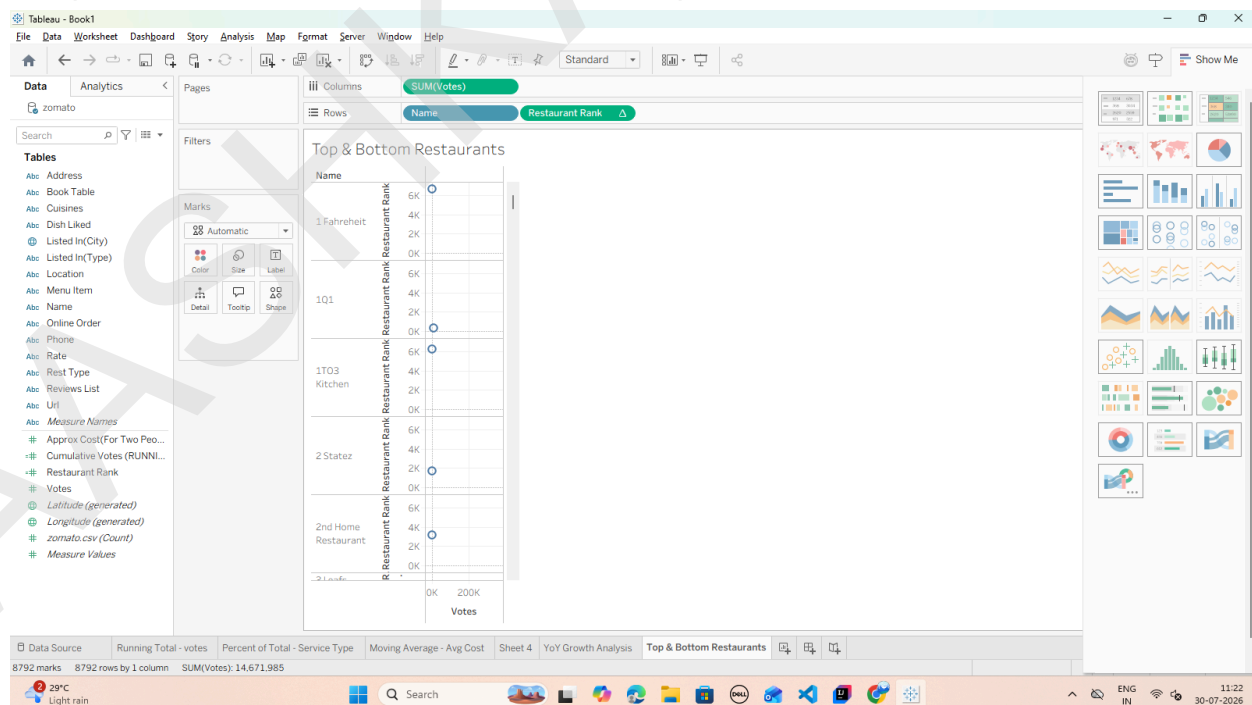
Task 1: Create a Rank Calculation

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.



Task 2: Build the Top N / Bottom N Ranked View

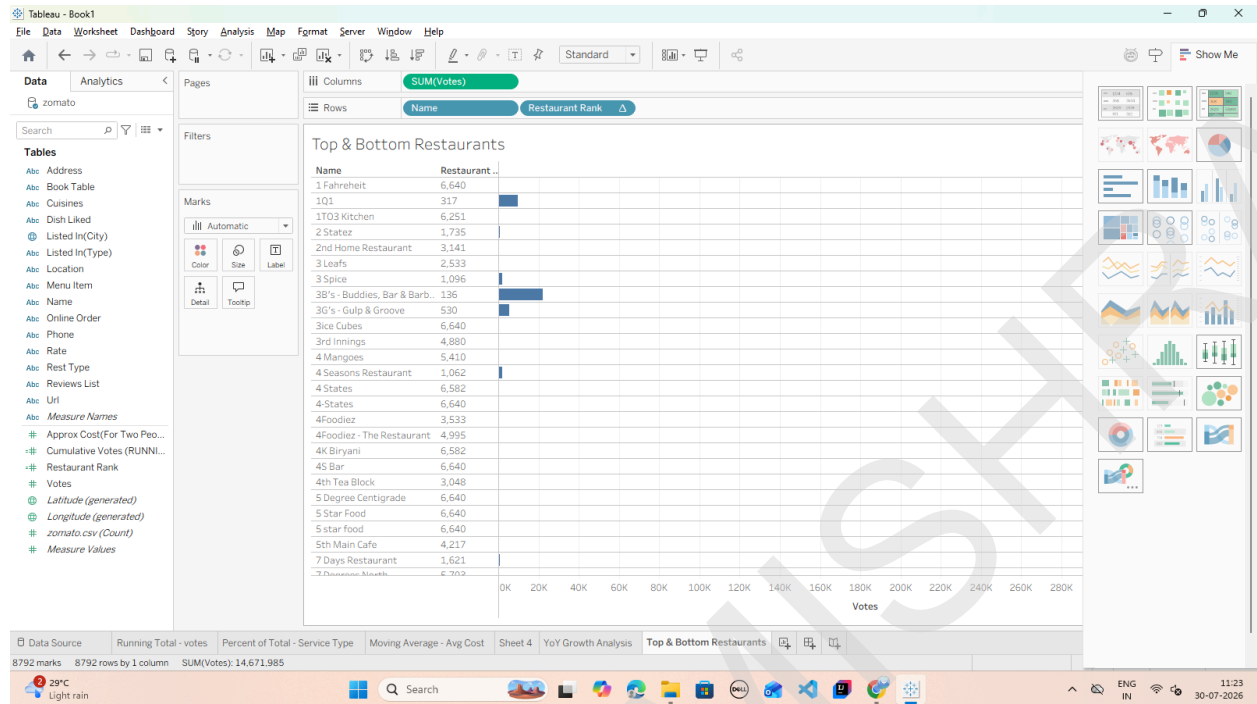
1. Open a new worksheet and rename it: Top & Bottom Restaurants.



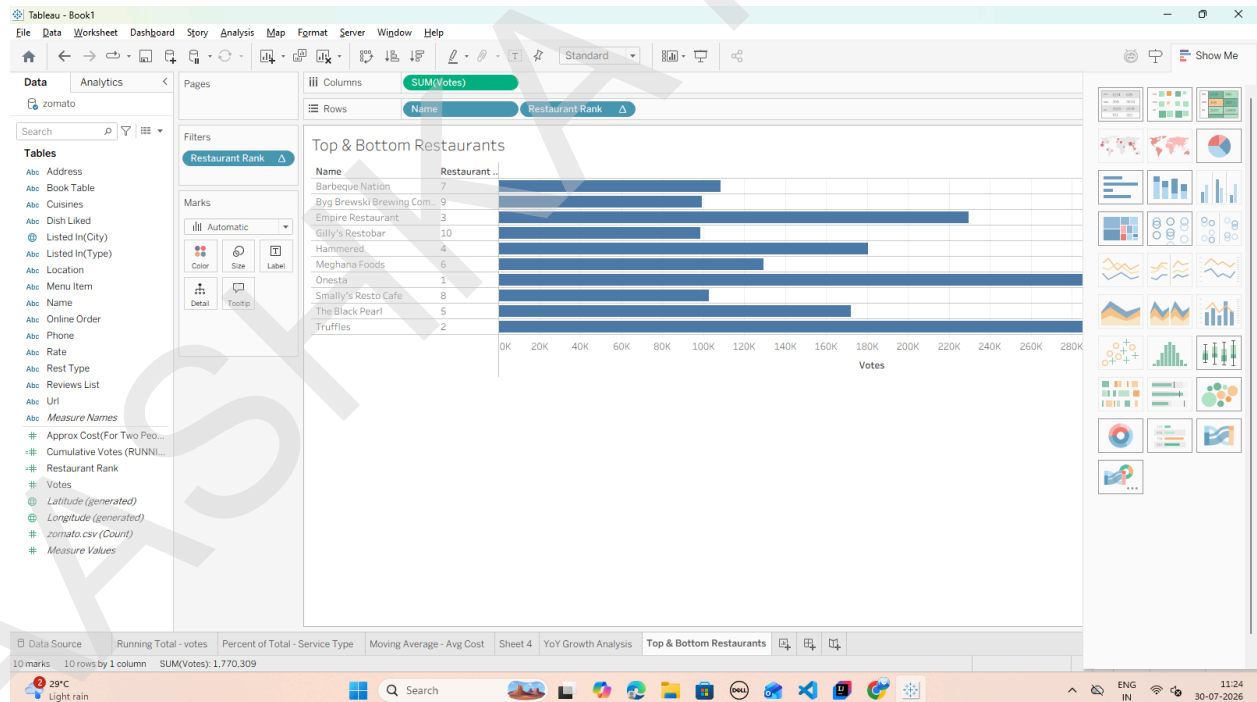
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## 2. Convert Restaurant Rank to a Discrete field:



## Task 3: Filter Dynamically to Display Top N Only



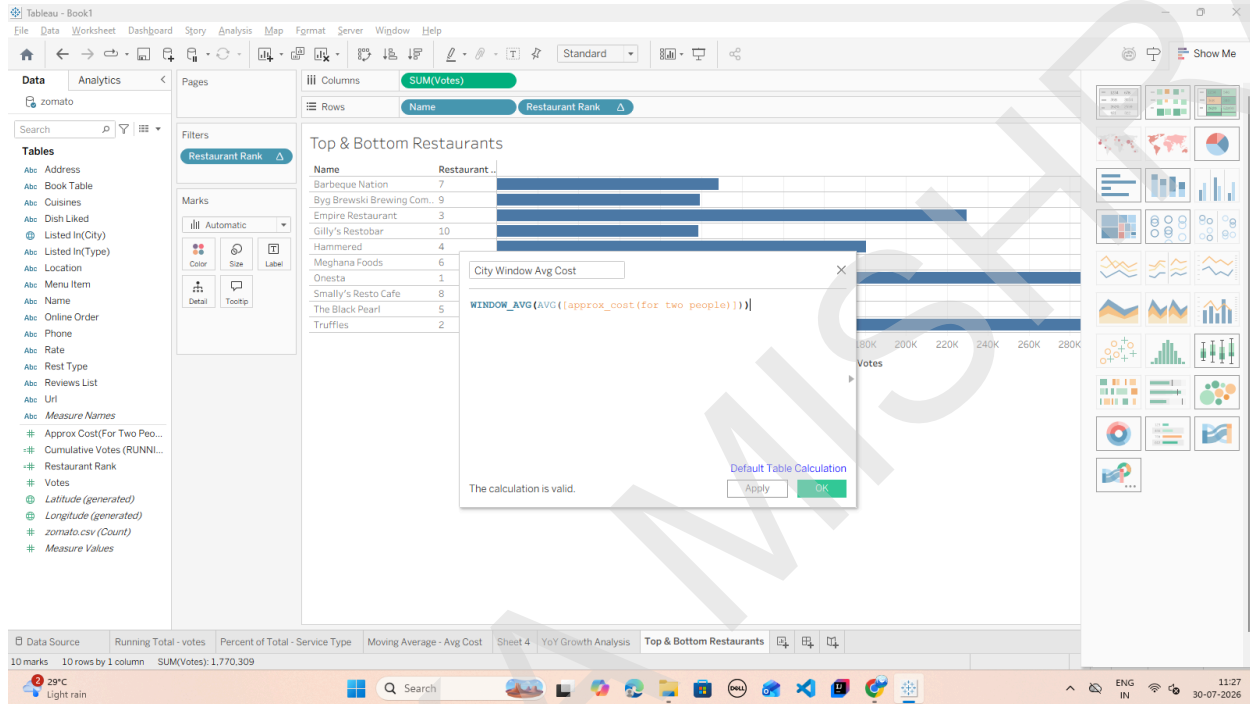
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**E: Implement window functions (WINDOW\_SUM, WINDOW\_AVG)**

**Task 1: Create a Window Average Calculation**

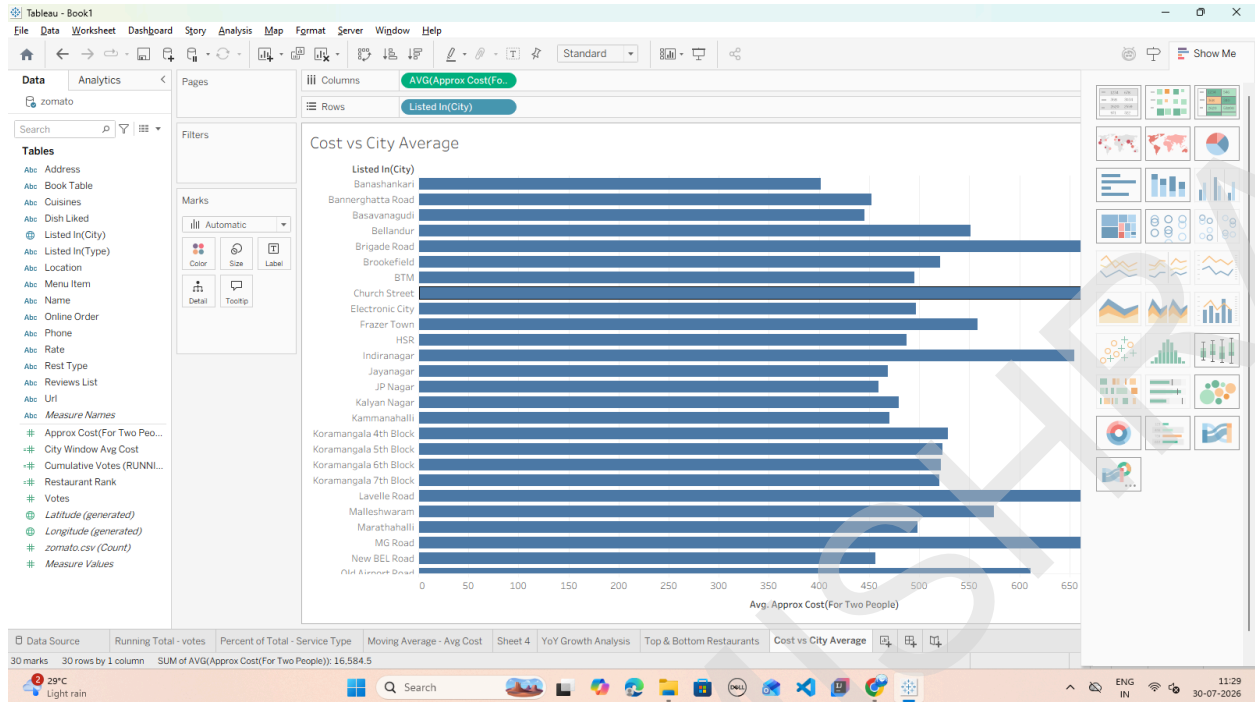
1. Click the small drop-down arrow at the top right of the Data Pane (next to the Search bar) and select Create Calculated Field.
2. Name the field: City Window Avg Cost.
3. Input the following window function formula: `WINDOW_AVG(AVG([approx_cost(for two people)]))`



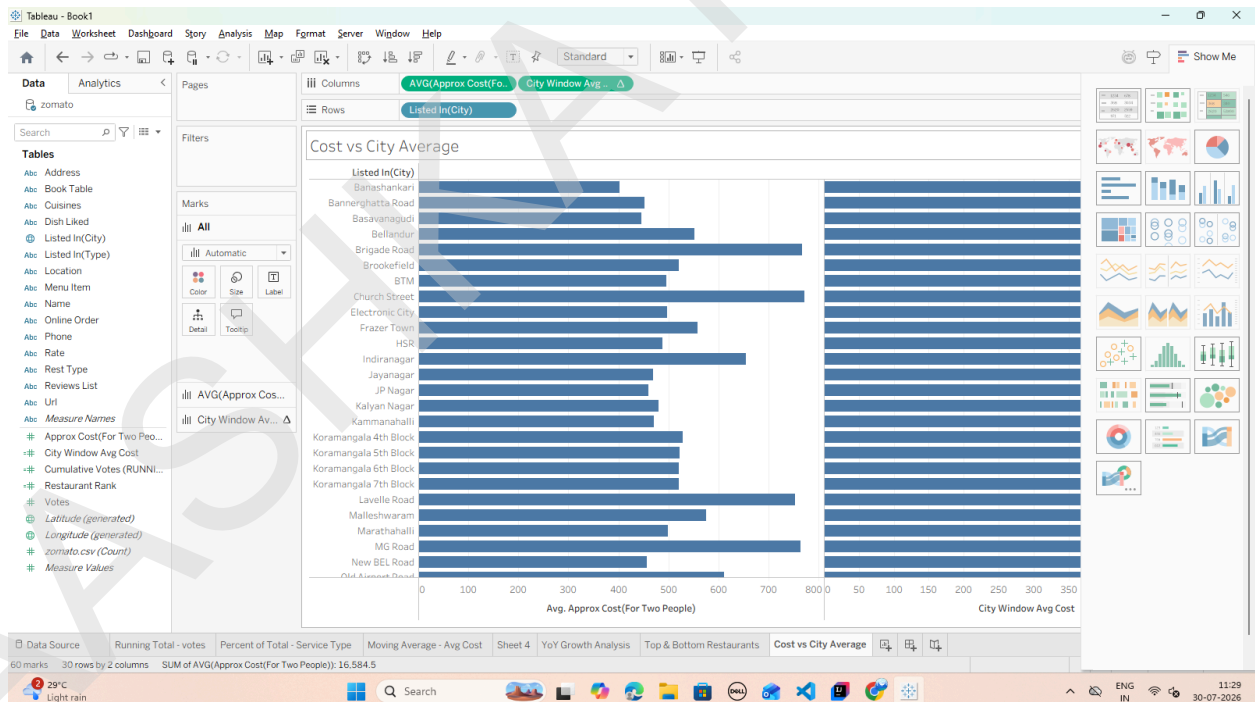
- Task 2: Build the Window Comparison Visualization**
1. Open a new worksheet and rename it: Cost vs City Average.
  2. Drag listed\_in(city) to the Rows shelf.
  3. Drag approx\_cost(for two people) to the Columns shelf. Change its aggregation to Average (Right-click pill  $\rightarrow$  Measure (Sum)  $\rightarrow$  Average).

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4. Drag your City Window Avg Cost calculated field onto the Columns shelf (placing it right beside AVG(approx\_cost...)).

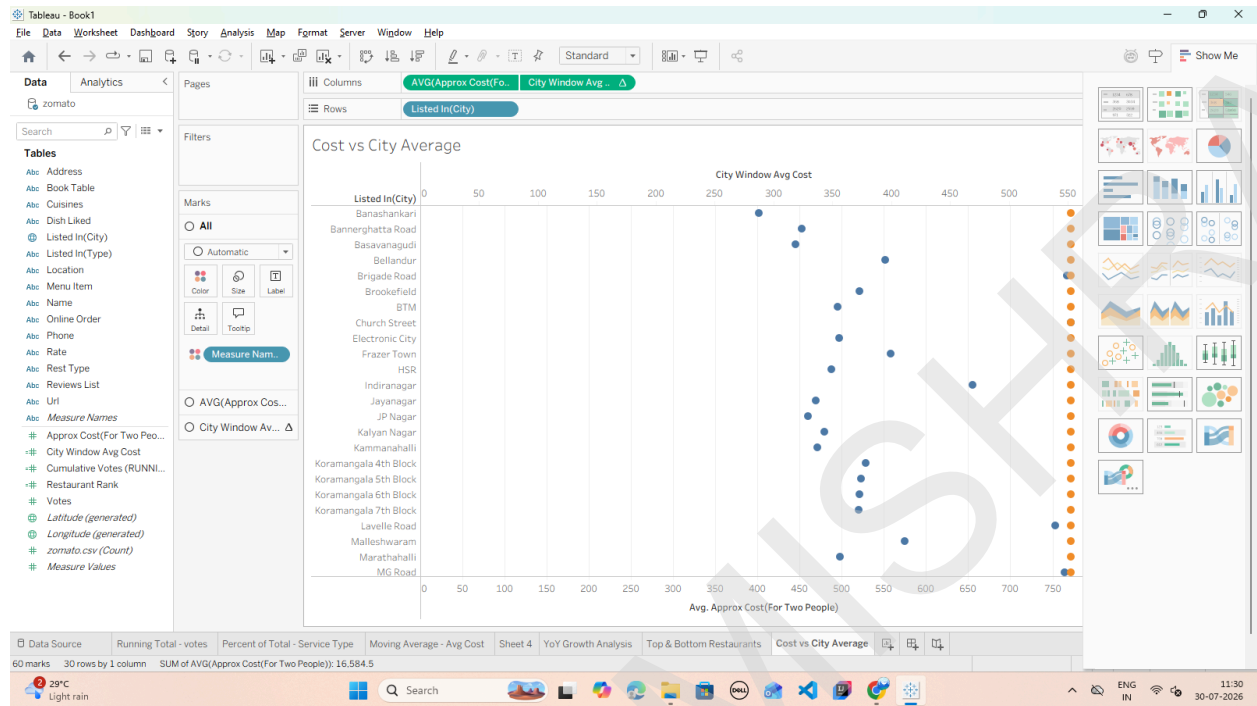


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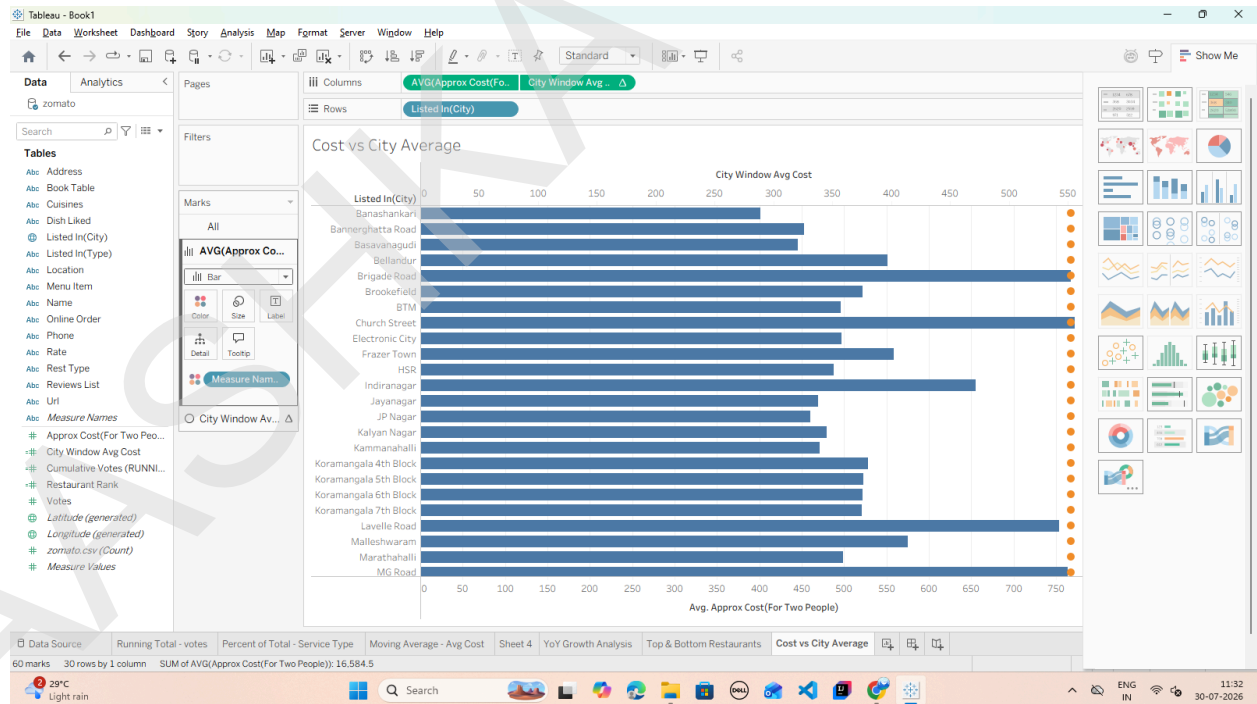
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**Task 3: Combine Measures into a Dual-Axis Reference View**

**1. Right-click the City Window Avg Cost pill on the Columns shelf and select Dual Axis.**



**3. In the Marks card: o Change the mark type for AVG(approx\_cost...) to Bar.**

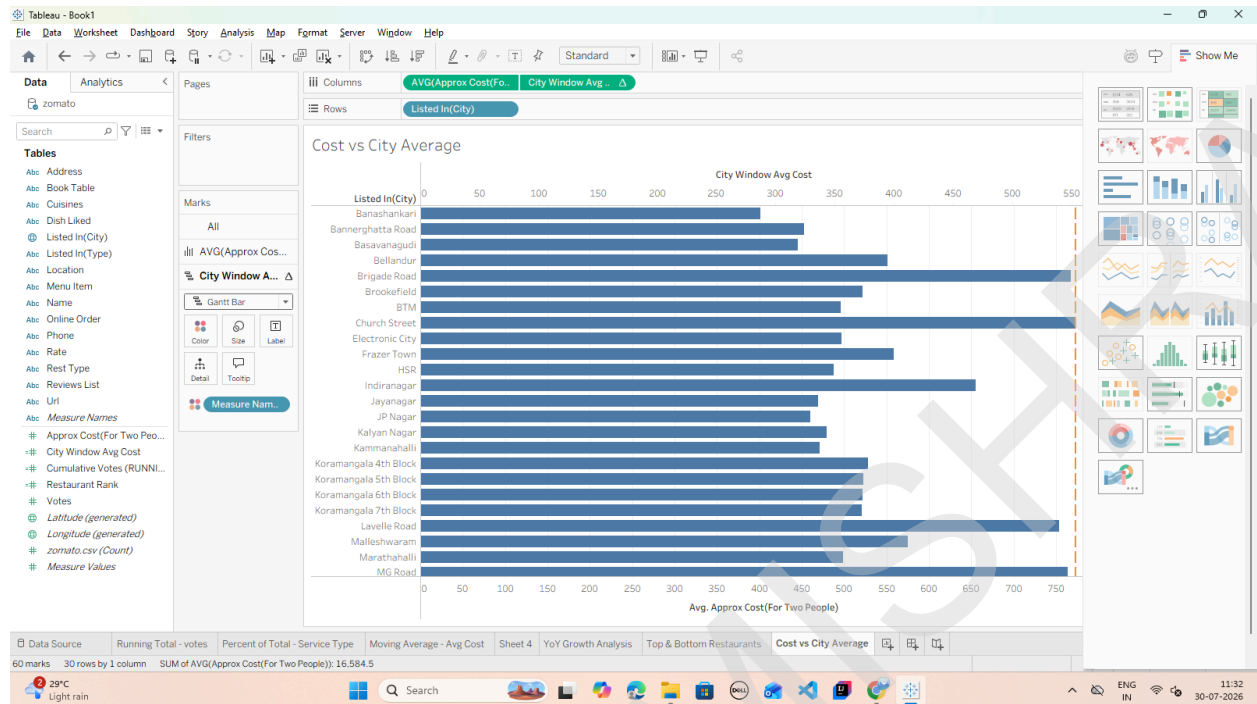


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o Change the mark type for City Window Avg Cost to Line (or Gantt Bar)

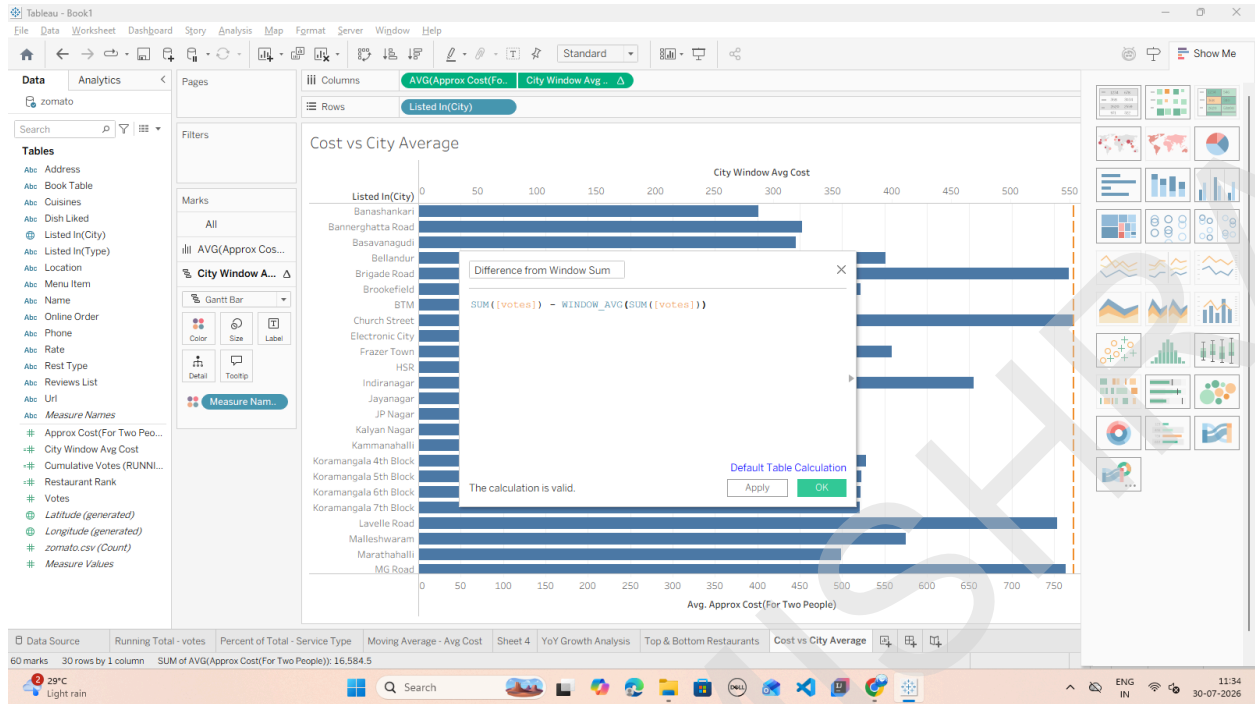


Task 4: Create a Window Difference Metric (WINDOW\_SUM)

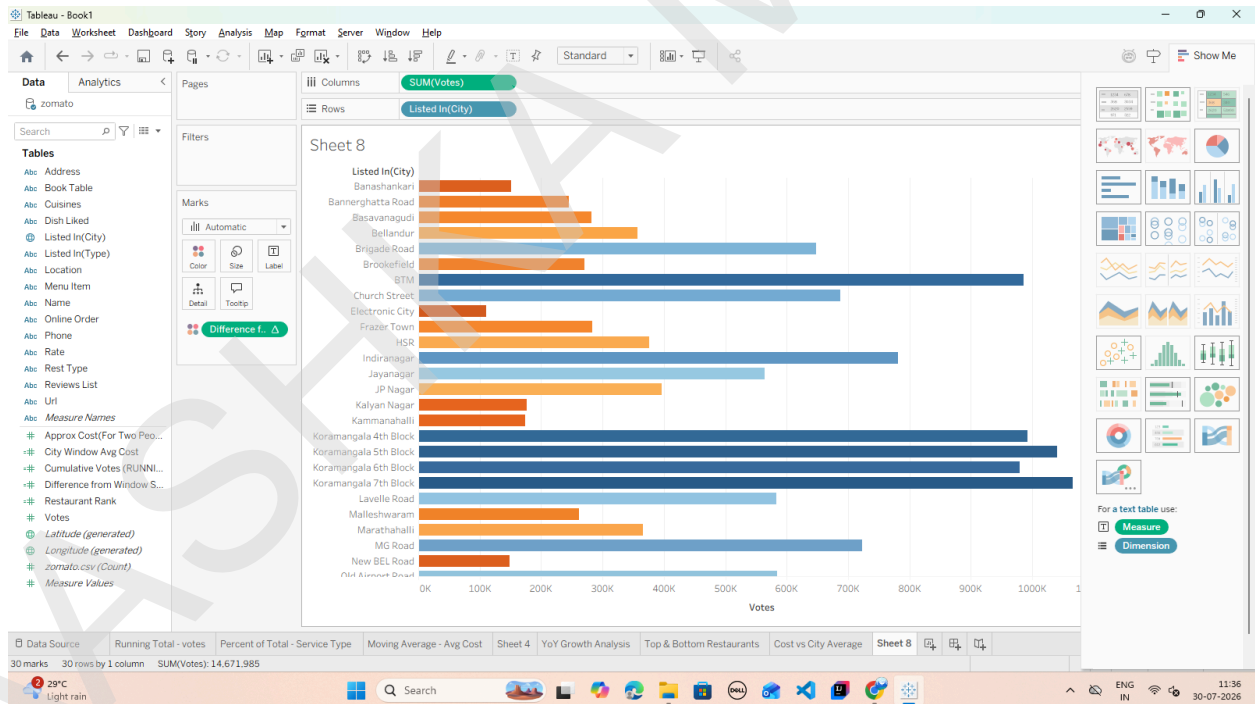
1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.
2. Name the field: Difference from Window Sum.
3. Input the following expression to compare each city's vote count against the total sum of all votes in the visible window:  $SUM([votes]) - WINDOW\_AVG(SUM([votes]))$

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### 4. Apply it to the View:



### F: Customize addressing and partitioning

#### Task 1: Build a Nested View for City and Service Type

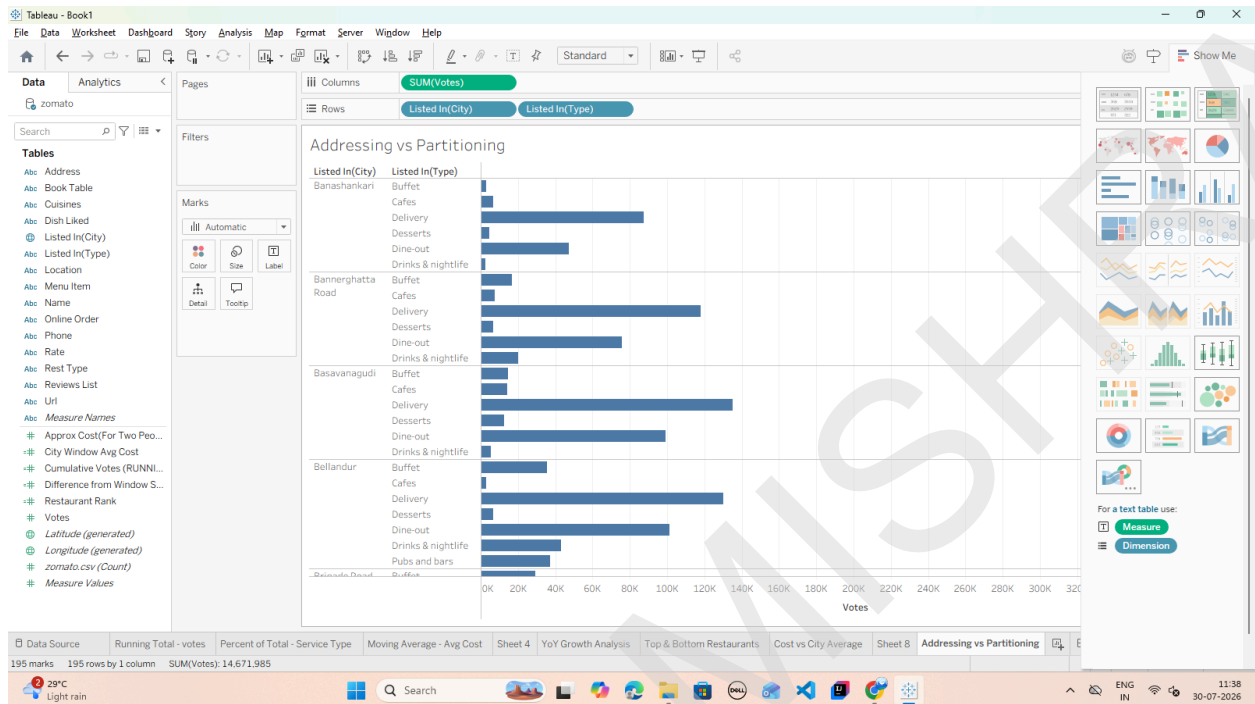
1. Open a new worksheet and rename it: Addressing vs Partitioning.
2. Drag listed\_in(city) to the Rows shelf.

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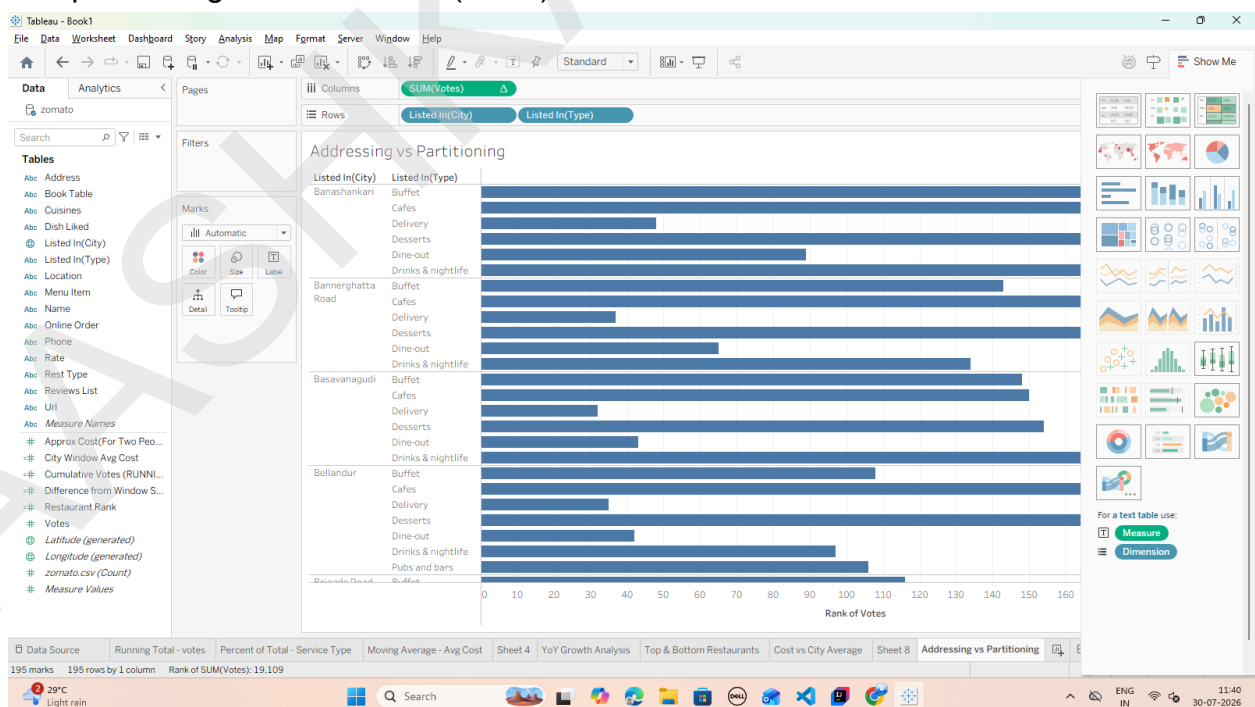
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3. Drag listed\_in(type) to the Rows shelf, placing it right next to listed\_in(city) (creating a two-level nested hierarchy).
4. Drag votes to the Columns shelf as SUM(votes).



**Task 2: Apply Partitioning via "Compute Using" Options**

1. Right-click the SUM(votes) pill on the Columns shelf, hover over Quick Table Calculation, and select Rank.
2. Right-click the SUM(votes) pill again and hover over Compute Using: o Select Table (down):

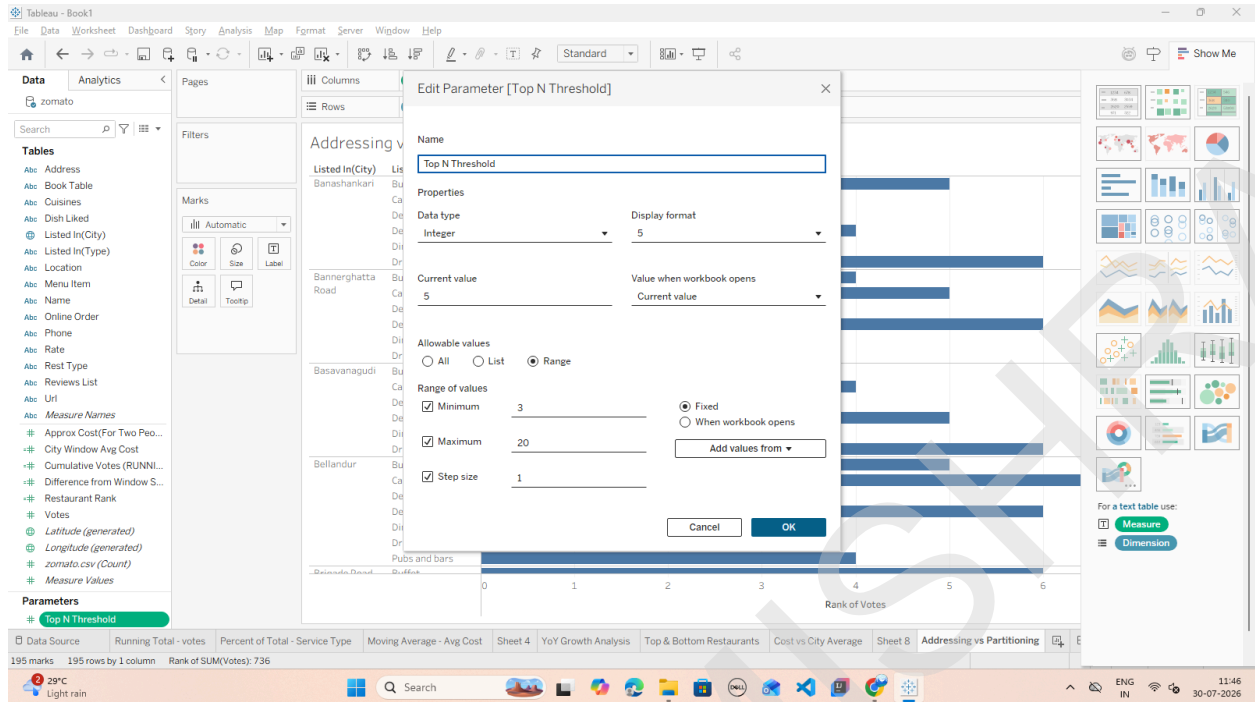


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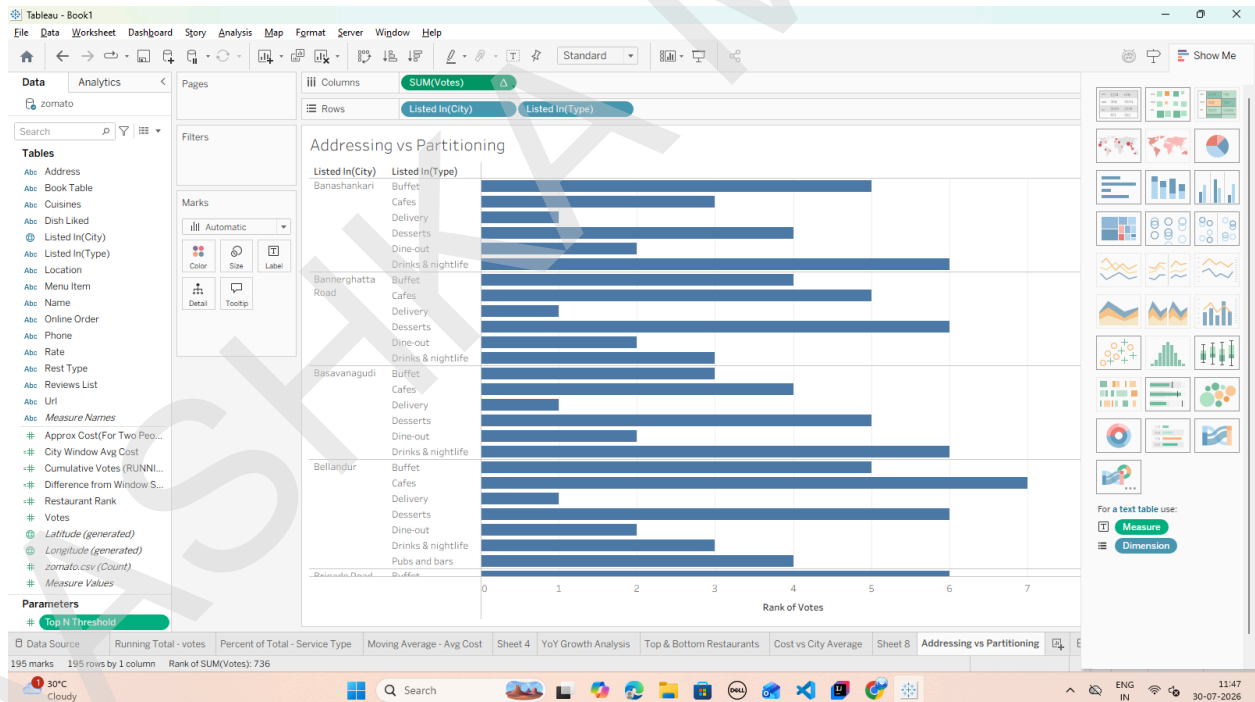


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### 5. Right-click Top N Threshold in the Data Pane and select Show Parameter Control.

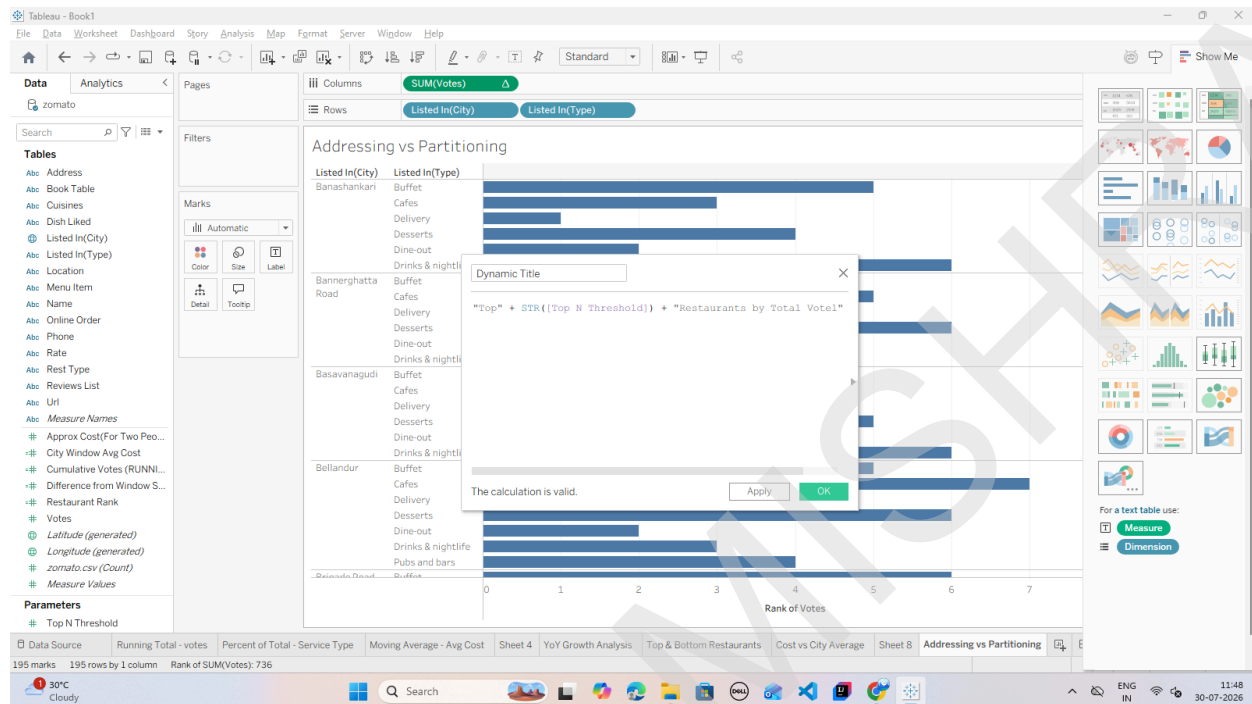


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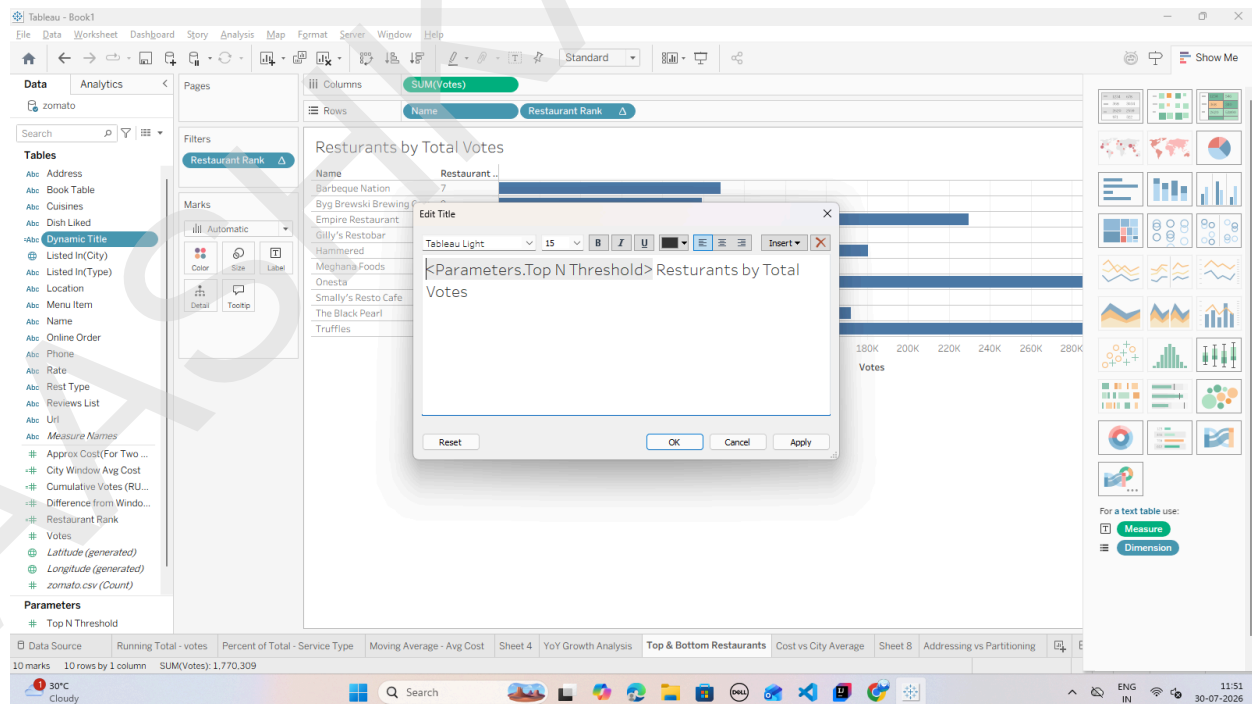
**Task 2: Create a Dynamic Header Calculation**

1. Create a new Calculated Field named Dynamic Title: "Top " + STR([Top N Threshold]) + " Restaurants by Total Votes"



**Task 3: Embed Calculated Fields and Parameters into the Sheet Title**

1. Double-click the Sheet Title at the top of your worksheet (e.g., Top & Bottom Restaurants).

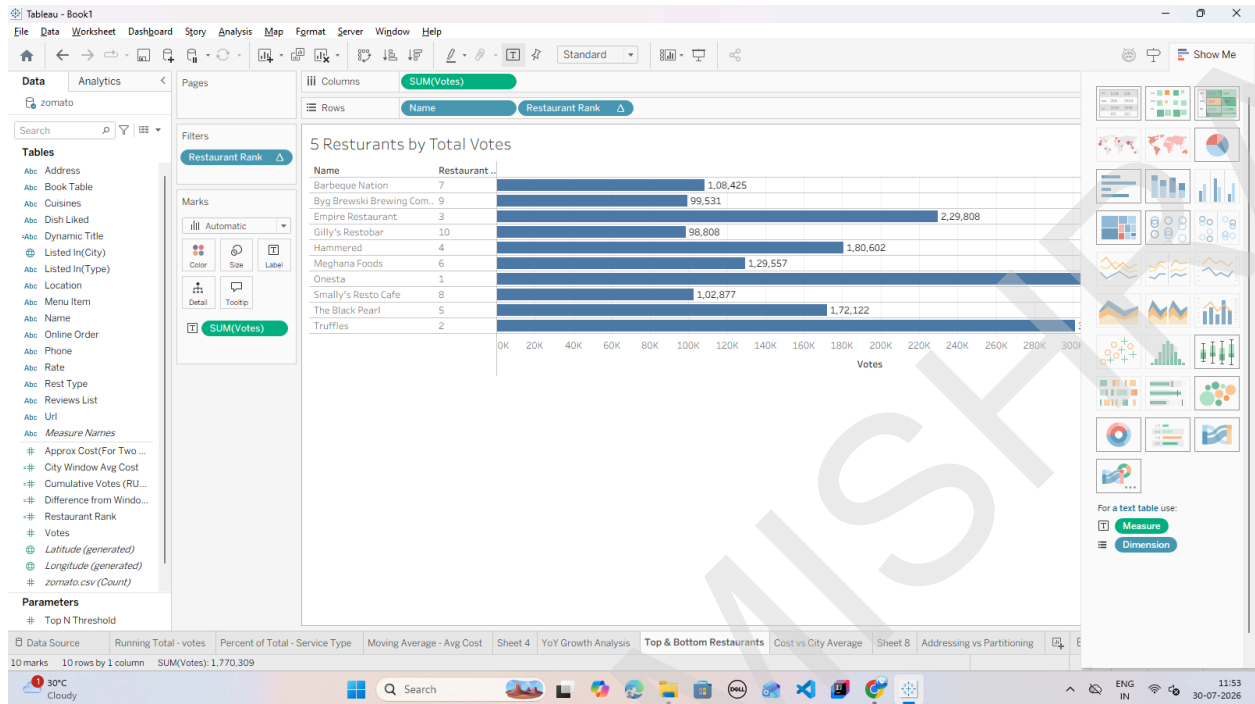


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Task 4: Add Dynamic Mark Labels (Highlighting Extremes)

1. Drag SUM(votes) to the Label card on the Marks card.



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